How GLP-1 Medications Are Changing the Caloric and Nutritional Content of Household Grocery Purchases

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Abstract

Growing use of GLP-1 medications for the treatment of a range of conditions has been speculated to induce big changes in food demand, with implications for nutrition and agricultural supply chains. We provide empirical evidence on this effect, drawing on transaction-level grocery data for a large panel of U.S. households merged with product-level nutrient contents. On a matched panel of adopter and never-adopter households, we use staggered difference-in-differences event studies to estimate how adoption of GLP-1 medications changes the calories and key nutrients embodied in grocery purchases. GLP-1 adoption reduces calories purchased by 5.1% in the adoption half-year, after which the calorie effect is no longer distinguishable from zero. Most nutrient contents of grocery purchases, including protein and fiber, decline along with calories at initiation, while sugar and added sugar do not fall significantly. Grocery spending falls by 7.1% in the adoption half-year. Across food groups, the most persistent cuts in both calories and spending occur in meat and beans. The biggest changes are made by adopters of weight-loss–indicated medications, who reduce calories purchased by 7.9% over the post-adoption horizon compared to an insignificant 1.1% for adopters of diabetes-indicated medications.

Researcher(s)' own analyses calculated (or derived) based in part on data from Nielsen Consumer LLC and marketing databases provided through the NielsenIQ Datasets at the Kilts Center for Marketing Data Center at The University of Chicago Booth School of Business. The conclusions drawn from the NielsenIQ data are those of the researcher(s) and do not reflect the views of NielsenIQ. NielsenIQ is not responsible for, had no role in, and was not involved in analyzing and preparing the results reported herein.

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1 Introduction

Glucagon-like peptide-1 receptor agonists (hereafter GLP-1) have rapidly become a popular class of medications for treating type 2 diabetes, obesity and overweight, and a potentially growing list of other human ailments (Drucker 2024). For weight management they have been shown in clinical trials to produce placebo-adjusted weight loss of 12 to 18%, exceeding any prior pharmacologic therapy (Jastreboff et al. 2022; Mozaffarian 2024). While the exact mechanisms are still under investigation, GLP-1 medications affect appetite and food intake, including by increasing satiety and reducing hunger, and consequently may alter both the quantity and composition of food demand (Martin et al. 2025; Zhu et al. 2025; Rutigliani et al. 2026).

By late 2025, roughly 18% of U.S. adults reported having ever used GLP-1 medications (Montero et al. 2025). Among adults who are not currently taking one and who report an overweight or obesity diagnosis, 43% said they would be interested, pointing to substantial room for further diffusion (Montero et al. 2025). Because they change how much and what people want to eat, the rapid spread of these medications has the potential to disrupt food systems, reshaping consumer demand for food, as well as the supply chains that provide it (Morgan Stanley Research 2026; Bartlow and Radtke 2026). Yet we know relatively little about how GLP-1 adoption changes the foods people eat, or about the caloric and nutritional consequences of those changes.

A nascent literature has begun to examine how GLP-1 adoption affects household food demand. Two studies use transaction-level data but stop short of measuring calories and nutrients. Hristakeva et al. (2026) find spending falls across most grocery categories, with item counts falling in proportion to the expenditure change. Awad et al. (2025) find budget shares shifting from ultra-processed toward minimally processed foods. These studies provide important evidence that GLP-1 adoption changes household grocery purchases. But item counts and processing level stand in for what households buy, and neither measures the calories or nutrients contained in the grocery basket.

Other evidence provides insight into changes in food intake and preferences among GLP-1 users. Dilley et al. (2025) ask survey respondents to report daily caloric intake over the previous week, and estimate that consumers taking GLP-1 medications for weight loss report consuming 720 to 990 fewer calories. Bina et al. (2026) estimate protein demand with a discrete choice model and a demand system, and find a higher willingness to pay for protein among GLP-1 users. However, these studies rely on recall and hypothetical choices rather than transactions. The empirical literature linking GLP-1 medications to food choices is still in its early stages. No study yet links actual household choices in food markets to their caloric and nutrient content. As a result, existing evidence cannot directly show how GLP-1 adoption changes the calories and nutrients contained in the food households actually purchase.

This distinction is important because reduced food intake can have different consequences for different nutrients. Clinical guidance has raised particular concerns about maintaining adequate protein intake during GLP-1 treatment. A joint advisory panel of four professional societies warns that GLP-1 therapy can erode muscle and bone along with fat (Mozaffarian et al. 2025). The same advisory panel emphasizes the importance of nutrition during treatment (Mozaffarian et al. 2025). Evidence also points to substantial losses of muscle among patients without nutrition counseling (Hoffman et al. 2026). More generally, changes in appetite may affect consumption of protein, fiber, sugar, fat, and other nutrients differently.

This paper examines how GLP-1 adoption changes the caloric and nutrient content of household grocery purchases. We address this question with a novel dataset that pairs a survey of GLP-1 use with a large consumer panel that records each household's grocery purchases at the product (Universal Product Code, UPC) level. We combine the NielsenIQ Consumer Panel with a GLP-1 survey administered to its panelists, about 40,000 households per wave. We then add each panelist's UPC-level grocery transactions, which record what households buy rather than what they report eating. Nutrient information comes from NielsenIQ, Syndigo, and the USDA FoodData Central database. By joining these purchase records to harmonized nutrition databases, we observe not only household grocery expenditures but also the calories and nutrients contained in the products households purchase.

This paper adds to the emerging literature linking GLP-1 medications to changes in food demand. Our primary contribution is to provide transaction-based evidence on how GLP-1 adoption changes the calories and nutrient content of the groceries that households purchase. Existing transaction-based studies measure expenditures, item counts, or broad food categories, while studies of food intake and nutrition generally rely on self-reported consumption or stated preferences. Our data allow us to measure directly what these approaches cannot: the calories and nutrients contained in actual household grocery purchases. We examine calories and several nutrients, including fiber, protein, sugar, fat, and sodium, and classify products by USDA ERS food groups (USDA-ERS 2016) to identify the food groups associated with changes in nutrient purchases. We also characterize how these effects evolve over time and differ across food groups and types of GLP-1 medications. Finally, our expenditure estimates provide an out-of-sample comparison with the emerging literature.

To estimate the effect of GLP-1 adoption, we match each adopting household to a never-adopting household on baseline demographics and a baseline diet-quality index. We then estimate dynamic treatment effects using a staggered difference-in-differences event-study design (Sun and Abraham 2021) with household and calendar half-year fixed effects. Adopting households enter treatment in different half-years, allowing us to trace changes in grocery purchases before and after adoption and

compare those changes with contemporaneous changes among matched never-adopting households.

We find that GLP-1 adoption initially reduces the caloric content of household grocery purchases. The decline is largest in the adoption half-year (5.1%) and subsequently attenuates. The nutrient response is not uniform. Fiber, protein, sodium, total fat, saturated fat, and total carbohydrate all fall significantly in the adoption half-year, with the effect fading over subsequent periods. The exceptions are sugar and added sugar, which show no significant decrease with GLP-1 adoption. Pooled over the post-adoption horizon, the reductions in fiber, protein, sodium, total fat, and saturated fat remain significant. Thus, the nutritional response cannot be characterized simply as a proportional reduction in all components of the grocery basket.

The food-group results help explain this pattern. In the adoption half-year, calories fall most sharply for meat and beans (11.6%), and this cut deepens to 16.0% two half-years out. Calories also fall significantly for grains, milk products, salty snacks, and desserts and sweets. The last of these does not persist, consistent with the absence of a sustained decline in sugar. The nutritional consequences of GLP-1 adoption depend on the changes in the composition of food purchases, not simply on the overall reduction in food purchases.

The calorie response is larger for households taking GLP-1 medications indicated for weight loss. Their calorie purchases fall by 7.9% over the post-adoption horizon, compared with a statistically insignificant 1.1% for diabetes-indicated medications. Households that later report stopping show no significant reduction in calories or spending.

These changes in calories and nutrients occur alongside a 7.1% reduction in grocery spending in the adoption half-year, consistent with the emerging evidence that GLP-1 adoption affects household food demand. The expenditure results provide economic context for the nutritional findings but, by themselves, cannot characterize their nutritional consequences. Together, the results show that GLP-1 adoption changes not only household grocery expenditures, but also the calories and nutrients contained in the food households purchase.

The remainder of the paper proceeds as follows. Section 2 describes the data and construction of the nutrition measures. Section 3 describes the empirical strategy. Section 4 presents the results on calories, nutrients, food groups, and grocery expenditures. Section 5 discusses the implications of the findings and concludes.

2 Data, Sample Construction, and Descriptive Statistics

2.1 Data

Our analysis combines survey, purchase, and nutrition data so that for each household we observe: (1) whether and when it adopts a GLP-1 medication, (2) what it buys in grocery markets, and (3) the caloric and nutrient content of those purchases. The NielsenIQ GLP-1 Panel Survey supplies the first, dating each household's adoption to a half-year. The NielsenIQ Consumer Panel supplies the second, recording every grocery transaction of those same households before and after adoption. Universal Product Code (UPC)-level Nutrition Facts data from NielsenIQ and Syndigo, augmented by the USDA FoodData Central database for non-barcoded and unlabeled items, supply the third by attaching calories and nutrients to each purchased product. No single source answers our question: adoption dates carry no information on diet, transactions carry none on nutrient content, and nutrient labels carry none on when a household started the medication. Our contribution comes from linking these three kinds of information, which yields a panel in which the caloric and nutrient content of a household's grocery purchases can be tracked across its own GLP-1 adoption date.

We track GLP-1 medication adoption and grocery purchases using an enhanced version of the NielsenIQ Consumer Panel combined with the NielsenIQ GLP-1 Panel Survey. Beginning at the end of 2023, the survey reports each household's self-reported GLP-1 use and the drug name. Each wave asks panelists whether they are currently taking a GLP-1 medication, whether they have taken one before, and which drug. The survey is fielded once per half-year, so a household contributes at most one use status per wave. We therefore work in half-years throughout, and adoption is dated no more finely than the half-year (Section 2.2 defines the adoption half-year). The drug name lets us classify use by the labeled indication, weight loss or diabetes management. Because panelists answer in every wave, we also observe use after adoption: a household is recorded as having stopped in the first later wave in which it reports no current use, and as a persistent user otherwise (Table 2). The survey covers approximately 40,000 households in each semiannual wave, or 73,344 unique households pooled across the 2024H1–2025H2 waves. The average current GLP-1 use rate is 12.8% over 2024H1–2025H2, rising from 9.9% in 2024H1 to 15.4% in 2025H2.

We combine the GLP-1 use data with household purchase records from the NielsenIQ Consumer Panel to examine how GLP-1 medications relate to grocery choices. The transaction-level records track each shopping trip, recording date, store, quantity, and expenditure for every item at the UPC level. Because the same households are followed over time, we observe their grocery baskets both before and after GLP-1 adoption, allowing us to measure within-household changes in grocery purchases. The data also include household characteristics such as income and education. Recording purchases at the UPC level is what lets us attach nutritional attributes to a household's shopping

basket, assigning every purchase to a food group and linking it to its nutrient content.

For each UPC purchased by a household, we combine the quantity purchased with product-level nutrition information to calculate the calories and nutrients contained in the household grocery purchases in each half-year. We draw that nutrition information from three sources. NielsenIQ supplies Nutrition Facts labels matched directly to the UPC, including calories, macronutrients (e.g., sugars and fats), and micronutrients (e.g., vitamins and minerals). Syndigo supplies the same label information for UPCs that NielsenIQ leaves unmatched. The USDA FoodData Central (FDC) database provides nutrient information for fresh produce that neither NielsenIQ nor Syndigo covers.

We merge nutrition values onto the purchase records and fill the fields that remain missing by imputation, in several steps. We first merge the data sets UPC by UPC, most specific source first, from NielsenIQ and then from Syndigo. We fill a blank nutrient with zero when its label reports other nutrition facts, because labeling rules require declaring any nutrient present in more than insignificant amounts (U.S. Food and Drug Administration 2024). Sugar is the exception, since its blanks less often mean absence. For fresh produce that no label covers, we take nutrient values for the same type of food from USDA FoodData Central. We fill the remaining blanks with the mean of similar products. After these steps, nutrient values cover 95.6 to 99.2 percent of grocery spending, depending on the nutrient. This follows approaches in the literature (Dubois et al. 2014; Cengiz and Rojas 2024); Appendix A.1 reports coverage for each major step. The estimated declines in calories and nutrients change little when we drop the imputed values (Appendix Table A4).

The linked data yield four sets of outcome variables, each measured for a household in a half-year. The first is calories purchased, the energy content of the household's grocery basket, in kilocalories. The second is the quantity of each of eight nutrients, in grams: dietary fiber, protein, total fat, saturated fat, sodium, total carbohydrate, sugar, and added sugar. The third is calories and expenditure in each of the eleven food groups defined below, in kilocalories and dollars. The fourth is total grocery expenditure, the household's spending on all food. The descriptive tables below express these quantities per adult equivalent, while the estimating equation in Section 3 uses the household total.

2.2 Sample Construction and Matching

To examine the potential impact on grocery purchases, we aggregate detailed UPC-level transactions into eleven broad groups based on both nutritional content and food type. Specifically, we adopt the food-group scheme of the USDA Economic Research Service (ERS) (USDA-ERS 2016), retaining six of the seven Tier-1 groups and disaggregating "Other Foods" into five of its Tier-2 subgroups, yielding eleven groups. Grounded in USDA nutritional standards, the scheme supports

analysis of nutrient intake across food groups (Zhen et al. 2024). We identify the food groups at the supercategory and category levels of the NielsenIQ classification and map them to the ERS scheme using the food-group definitions in USDA-ERS (2016). Non-food items such as household supplies are excluded. We aggregate the transaction data to the household by half-year level, both in total and for each food group.

Our goal is to estimate the average treatment effect on the treated (ATT) of GLP-1 adoption on grocery purchases. The primary empirical challenge is that GLP-1 adoption is not randomly assigned: households that adopt GLP-1 medications may differ systematically from non-adopters in ways that also affect grocery purchases. To improve the comparability of GLP-1-using and non-using households, we first match households on observed characteristics. We then estimate the ATT with a difference-in-differences framework, detailed in Section 3.

We use grocery purchase records from 2023 through 2025. We define treated households as those whose first recorded GLP-1 use falls in 2024H1–2025H2, and control households as households that report no GLP-1 use. An adopter's adoption half-year is the first wave in which it reports using a GLP-1 medication, provided it reported non-use in an earlier wave. Households first reporting use in the initial 2023H2 wave therefore cannot be dated. Each adopter must also be observed at least one half-year before adoption and one in or after its adoption half-year. We keep a household's half-year only if the household records purchases in at least five of its six months, so that changes in recorded purchases reflect behavior rather than gaps in reporting.

We employ a balanced one-to-one matching protocol. Specifically, each GLP-1-adopting household is paired, without replacement, with the nearest never-adopting household by Mahalanobis distance on a vector of pre-treatment characteristics. We restrict matches to within a caliper of 0.1 standard deviations of the estimated propensity score (Rosenbaum and Rubin 1985; Imbens and Rubin 2015). We use the following variables as matching covariates: income, age, an indicator of children in the household, Supplemental Nutrition Assistance Program (SNAP) participation, and a baseline diet-quality index. Matching on the Mahalanobis distance balances each covariate directly, while the propensity-score caliper enforces common support.

The baseline diet-quality index is a single standardized score that summarizes a household's pre-adoption dietary composition. We use four nutrients: added sugar, saturated fat, sodium, and dietary fiber. For each household, we compute the energy-normalized density of each nutrient (per 1,000 kcal) over its purchases in its earliest one or two pre-adoption half-years. We then standardize each density to a z-score across households. We also reverse the sign of the fiber z-score so that larger values denote a less healthy diet, and average the four standardized values into a single index.

After matching, the treatment and control groups are balanced on all matching covariates (Table 1, Panel A). The matched sample comprises adopting households whose first reported use falls in

2024H1 through 2025H2, each paired one-to-one with a never-adopting household, for a total of 2,128 matched pairs.

Panel B of Table 1 summarizes the outcome variables over the estimation sample, separately for adopting and never-adopting households and including half-years after adoption. Quantities are per adult equivalent per half-year, so they describe what one adult brings home over six months. Across the two groups, the average household–half-year records 316,000 to 337,000 calories purchased and $1,546 to $1,761 of grocery spending. Several outcomes have a standard deviation above the mean. Added sugar is the extreme case, with a mean of 22,502 grams and a standard deviation of 38,010 among never-adopting households. These levels give the base that turns the percentage effects reported below into quantities.

Table 2 describes the four adoption cohorts, each the set of adopters whose first reported use falls in the same half-year. Cohort sizes range from 359 to 642 households. One half-year after a cohort starts, between 54% and 74% of its households still report using GLP-1, while 17% to 22% report having stopped. The remaining households give no survey response in that half-year. Their share is larger for later-adopting cohorts, whose follow-up wave falls closer to the end of the survey window. When every subsequent wave is counted rather than only the first, 19.2% of adopters in the estimation sample report stopping at some point. We use that information below to separate households that stay on treatment from those that do not.

2.3 Descriptive Statistics

The eleven food groups are summarized in Tables 3 and 4. Throughout the descriptive tables that follow, we report means over the estimation sample; a GLP-1 user household enters the pre-initiation or initiation column whenever it is observed in that half-year. Comparisons between the pre and initiation columns pool cohorts observed at different calendar times, so we defer causal statements to the event-study estimates. Adult equivalents are the sum of household members' daily energy requirements divided by 2,245 kcal (HMSO 1991).

Table 3 reports average purchased calories (in kilocalories) per adult equivalent for total food and for each of the eleven food groups. GLP-1 user households purchase more calories than their matched controls in most food groups before first use. The largest differences are in calorie-dense, processed groups such as prepared meals, desserts and sweets, and salty snacks. Upon first use, calories fall in table fats and oils, meat and beans, and prepared meals but hold up in salty snacks and fruit.

Table 4 reports expenditure (in real dollars) per adult equivalent. On average, GLP-1 user households spend less on groceries per adult equivalent in the half-year of first use, and expenditure differences relative to control households vary considerably across food groups. Before initiation,

Table 1: Covariate balance after matching and descriptive statistics of outcome variables

	Non-adopter		GLP-1 adopter			
	Mean	SD	Mean	SD	Diff	P-value
Panel A. Covariate balance after matching						
Number of households	2,128		2,128			
Income < $125,000	0.745	0.436	0.746	0.436	0.000	0.972
Age < 65	0.820	0.384	0.818	0.386	-0.003	0.811
Has Children	0.126	0.332	0.128	0.334	0.001	0.890
SNAP Current Participant	0.102	0.302	0.102	0.302	0.000	1.000
Baseline Diet Index	0.088	0.386	0.089	0.387	0.001	0.965
Panel B. Outcome variables (per adult equivalent per half-year)						
Calories (thousand kcal)	315.8	263.3	337.1	288.9		
Total fat (g)	11,970	8,954	13,318	10,746		
Saturated fat (g)	4,155	3,134	4,570	3,648		
Total carbohydrate (g)	46,788	50,871	48,550	53,104		
Sugar (g)	27,691	39,854	27,713	41,342		
Added sugar (g)	22,502	38,010	22,547	40,132		
Dietary fiber (g)	1,902	1,664	2,057	1,743		
Protein (g)	7,437	5,664	8,029	5,879		
Sodium (g)	482.2	447.7	532.4	502.2		
Grocery expenditure (real USD)	1,546	1,500	1,761	1,920		

Notes: Matched sample from 1:1 nearest-neighbor Mahalanobis matching within a caliper of 0.1 standard deviations of the estimated propensity score, without replacement (Rosenbaum and Rubin 1985 hybrid). Means and standard deviations; difference is treated minus control. P-values are from two-sample t-tests. The baseline diet index is a z-scored composite of added sugar, saturated fat, sodium, and reverse-signed fiber densities; higher values denote a less healthy diet. The table describes the matched sample, which is also the estimation sample. See the appendix table on cohort composition. Panel B describes the outcome variables for each group over the estimation sample: every household–half-year entering the event-study regressions (4,256 households over 6 half-years, 23,078 observations), including half-years after adoption. Differences between the group means are not estimates of treatment effects; those are reported in Section 4. Quantities are per adult equivalent per half-year, on the HMSO (1991) energy scale; expenditure is in real dollars. *Source:* Authors' calculations using the NielsenIQ Consumer Panel.

Table 2: Adoption Cohorts: Size, Continued Adoption, and Attrition

Adoption cohort	Households	Continued adoption (%)	Attrition (%)	No follow-up (%)
2024H1	359	74.4	17.0	8.6
2024H2	642	60.7	22.0	17.3
2025H1	564	54.3	21.6	24.1
2025H2	563	—	—	—

Notes: Each row is a half-year adoption cohort, defined by the first half-year in which a household reports GLP-1 use in the NielsenIQ GLP-1 survey. Households are the treated households matched in the estimation panel. *Continued adoption* is the share of the cohort still reporting GLP-1 use one half-year after adoption; *attrition* is the share reporting that they are no longer using GLP-1; *no follow-up* is the share with no GLP-1 status recorded in that half-year, the survey being an unbalanced panel. All three use the full cohort as the denominator and sum to 100. The 2025H2 cohort adopts in the final half-year, so none of the three rates is defined for it. Cohorts adopting later have a higher no-follow-up share because their follow-up half-year sits closer to the end of the survey window. *Source:* Authors' calculations using the NielsenIQ Consumer Panel and GLP-1 survey.

user households also spend more than controls on most food groups. Grocery expenditure declines across all groups upon first use, with the decline most pronounced in table fats and oils, prepared meals, and meat and beans.

GLP-1 user households purchase about 335,700 kcal per adult equivalent in the half-year before first use and 317,400 kcal in the half-year of first use. Spending falls over the same two half-years, from $1,766 to $1,646 per adult equivalent. Expenditure per 1,000 kcal for all food falls from $5.07 to $5.02 across the two half-years. These are ratios of pooled totals, not of the per-adult-equivalent means in Tables 3 and 4. Appendix Table A5 reports the food-group detail.

Each nutrient's food-group sources govern how a spending change in any one group passes through to the nutrient totals. Protein is spread across several groups, with meat and beans (26%) and milk products (21%) the largest, while beverages alone supply 58% of total sugar, so a given percentage cut in beverage spending moves sugar more than the same cut anywhere else (Appendix Tables A6 and A7).

3 Identification

We identify the ATT of GLP-1 adoption on grocery purchases in two steps: matching, then difference-in-differences on the matched sample. First, matching balances adopting and never-adopting households on observable baseline characteristics (Section 2.2), and Table 1 reports the resulting balance. Second, we estimate the difference-in-differences by pooling all matched households into a single event-study regression. Identification comes from the staggered timing of first GLP-1 use across households. Adopting households enter treatment in different half-years,

Table 3: Half-year purchased calories per adult equivalent by food group

Category	GLP-1 User Households		Controls
	Pre (6 months)	Initiation (6 months)	
	(kcal per adult equivalent)		
Number of households	2,128	2,128	2,128
All Food	335,675	317,355	315,844
Beverages	72,505	69,347	72,685
Desserts, sweets, and candies	50,354	47,797	47,117
Fruit	5898	6083	6264
Grains	33,844	32,479	33,606
Gravies, sauces, condiments, and spices	11,742	10,812	11,118
Meat and beans	32,996	29,931	30,563
Milk products	32,352	30,871	30,840
Prepared meals, sides, and salads	39,741	35,941	32,359
Salty snacks	32,233	32,591	29,243
Table fats, oils, and salad dressings	21,145	19,506	19,914
Vegetables	7367	6988	7052

Notes: Each cell reports the mean half-year purchased calories per adult equivalent (kcal). The sample is the estimation sample, the same sample as the event-study regressions; a household enters a column whenever it is observed in the corresponding period. Calories are in kilocalories (kcal). Adult equivalent: daily caloric needs by age group (HMSO 1991) summed across household members and divided by 2,245 kcal. Column 1: same user households in the half-year before first GLP-1 use (pre-treatment); column 2: 2024H1–2025H2 GLP-1 user households in the half-year in which their first GLP-1 use falls (initiation half-year); column 3: 1:1 matched controls, averaged over all half-years in the estimation window rather than at a single event time; the within-adopter change across columns 1–2, not the gap with column 3, is the comparison of interest. "Number of households" reports the count of distinct households in the estimation sample for each group. The household count is the number of distinct households in that estimation sample. *Source:* Authors' calculations using the NielsenIQ Consumer Panel.

Table 4: Half-year expenditure per adult equivalent by food group

Category	GLP-1 User Households		
	Pre (6 months)	Initiation (6 months)	Controls
	(real USD per adult equivalent)		
Number of households	2,128	2,128	2,128
All Food	1766.37	1646.46	1545.71
Beverages	454.08	417.50	366.98
Desserts, sweets, and candies	223.28	217.91	194.49
Fruit	60.80	60.67	62.76
Grains	111.82	103.59	103.45
Gravies, sauces, condiments, and spices	74.14	68.44	71.04
Meat and beans	220.91	203.01	204.17
Milk products	163.13	157.87	152.20
Prepared meals, sides, and salads	225.85	206.01	188.29
Salty snacks	111.44	105.18	96.80
Table fats, oils, and salad dressings	44.21	38.30	39.13
Vegetables	100.07	93.39	90.74

Notes: Each cell reports the mean half-year expenditure per adult equivalent (real USD). The sample is the estimation sample, the same sample as the event-study regressions; a household enters a column whenever it is observed in the corresponding period. Expenditure is in real dollars. Adult equivalent: daily caloric needs by age group (HMSO 1991) summed across household members and divided by 2,245 kcal. Column 1: same user households in the half-year before first GLP-1 use (pre-treatment); column 2: 2024H1–2025H2 GLP-1 user households in the half-year in which their first GLP-1 use falls (initiation half-year); column 3: 1:1 matched controls, averaged over all half-years in the estimation window rather than at a single event time; the within-adopter change across columns 1–2, not the gap with column 3, is the comparison of interest. "Number of households" reports the count of distinct households in the estimation sample for each group. The household count is the number of distinct households in that estimation sample. *Source:* Authors' calculations using the NielsenIQ Consumer Panel.

while the never-adopting households are untreated throughout. The estimate is the change in adopters' purchases around their own adoption dates, net of the contemporaneous change among never-adopting households.

The key identification assumption for this framework is the parallel-trends assumption. In the absence of GLP-1 adoption, the grocery purchases of adopting households would have followed the same trend as those of the matched never-treated households.

Three threats to this parallel-trends assumption deserve particular attention. The first is time-varying confounding, whereby GLP-1 adoption may be triggered by household-specific health shocks or other contemporaneous events that also affect grocery purchasing. The second is that households may already be on a distinct purchase trajectory in the periods immediately preceding adoption. Matching on baseline levels does not make such a path parallel to that of comparison households. The third is anticipation. The decision to adopt is preceded by physician visits and the filling of a prescription, so households may alter their diet before the half-year in which adoption is recorded. That would contaminate the reference period, the half-year immediately preceding recorded adoption, and bias the estimated dynamic effects.

We address these concerns through four complementary strategies, reported in full in the results section and appendix. First, we match on baseline diet quality and demographics, improving the comparability of treated and control households and making the parallel-trends assumption more plausible. Second, we inspect the event-time coefficients for the half-years before adoption, looking for differential pre-trends. Third, we bound the sensitivity of the estimates to parallel-trends violations using the honest difference-in-differences approach of Rambachan and Roth (2023). Fourth, we probe whether anticipation or coarse half-year adoption timing drives the results.

We estimate the following Sun and Abraham (2021) specification:

$$\log(1 + \text{Grocery}_{ih}) = \sum_{e \in \mathcal{E}, e \neq -1} \beta_e \mathbb{1}\{h - g_i = e\} + \alpha_i + \lambda_h + \varepsilon_{ih} \tag{1}$$

where the dependent variable, $\log(1 + \text{Grocery}_{ih})$, is the log-plus-one transform of household i's total grocery purchases during half-year h. We estimate the same dynamic treatment-effect framework for a family of outcomes describing the economic and nutritional content of household grocery purchases. The outcomes are calories, the quantity of each of eight nutrients (dietary fiber, protein, total fat, saturated fat, sodium, total carbohydrate, sugar, and added sugar), calories and expenditure in each of eleven food groups, and total spending in dollars.[1] We estimate the model separately for

[1]The outcome is the household total. The household fixed effects absorb each household's average size. Only within-household size changes over our three-year window, uncommon in this panel, could shift totals without a per-member behavioral change.

each outcome.

Adding one before taking logs keeps the half-years in which a household records zero purchases. Such zeros are common for the disaggregated food-group outcomes and rare for the household totals. The coefficients approximate percentage changes, exact only in the small-change limit.

In Equation 1, g_i is the half-year in which household i first adopted GLP-1 medications, $e = h - g_i$ is the event time in half-years, and $\mathcal{E}$ collects the event times present in the sample. The coefficients β_e measure the average effect of GLP-1 adoption e half-years relative to adoption, with the period immediately preceding adoption ($e = -1$) serving as the omitted reference. We also report a single pooled ATT, the observation-weighted average of the β_e for $e \geq 0$.

Because the survey dates adoption only to the half-year, $e = 0$ is the half-year that contains a household's first recorded use, not the six months following it. Exposure within that period runs from zero to six months, roughly three on average. We therefore read β_0 as the effect of a partially treated half-year, which understates the effect of a fully treated one. We refer to it throughout as the adoption half-year.

The model includes two sets of fixed effects. Household fixed effects (α_i) absorb all time-invariant household characteristics that could correlate with both GLP-1 uptake and grocery shopping, such as long-term dietary preferences and baseline health consciousness. Half-year fixed effects (λ_h) absorb aggregate time shocks and seasonality, such as food-price fluctuations and macroeconomic conditions. We cluster standard errors at the household level to allow for arbitrary serial correlation within a household. Clustering at the matched-pair level instead leaves inference essentially unchanged (Appendix C.4).

We adopt the Sun and Abraham (2021) interaction-weighted estimator as our primary specification. It corrects for the negative-weighting bias of two-way fixed effects (TWFE) under heterogeneous treatment effects. It also delivers dynamic treatment effects with the same flexibility as TWFE. Robustness to alternative estimators, including standard two-way fixed effects, Callaway and Sant'Anna (2021), Borusyak et al. (2024), and Roth and Sant'Anna (2023), is documented in Appendix Figure A1.

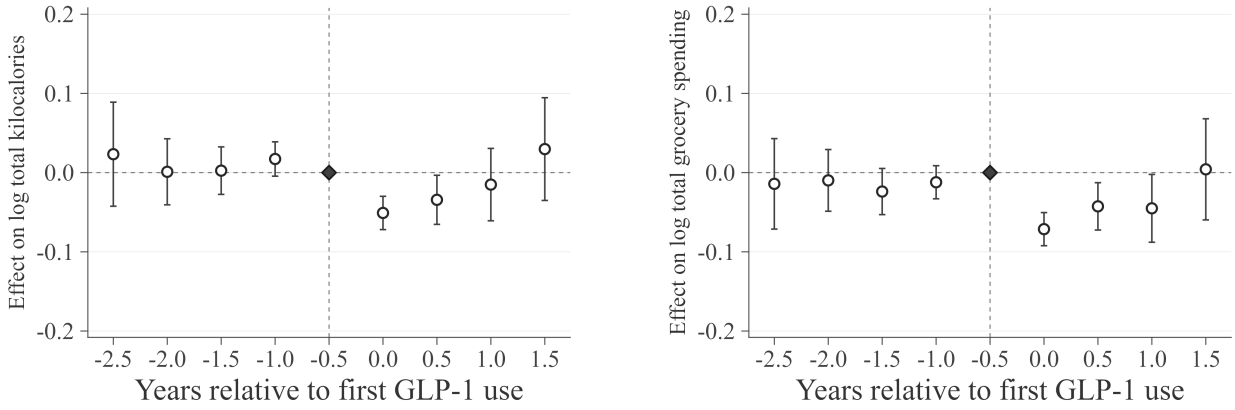
4 Results

We estimate the effect of GLP-1 adoption on calories purchased and total grocery spending, then on eight nutrient aggregates. We explore how those aggregate effects break down across eleven food groups and medication types. The two totals and the eight nutrient aggregates are our primary outcomes. They are a small set of hypotheses fixed in advance, to which we attach a causal

interpretation and the sensitivity analysis reported in the Robustness subsection.

4.1 Calories and Spending

Figure 1 plots event-study coefficients for calories purchased (panel a) across all half-years relative to GLP-1 adoption. In the adoption half-year, GLP-1 adoption reduces calories purchased by 5.1% ($p < 0.001$). The caloric effect then fades: -3.4% ($p = 0.03$) one half-year out and an insignificant -1.5% ($p = 0.52$) two half-years out. Averaged over the post-adoption horizon, calories fall by 3.4% ($p = 0.006$). Pre-adoption calorie coefficients are jointly indistinguishable from zero ($p = 0.28$; Section 4.5).



(a) Total calories purchased

(b) Total grocery spending

Figure 1: Effect of GLP-1 adoption on total grocery outcomes, by event time

Notes: Coefficients are from the Sun and Abraham (2021) staggered difference-in-differences estimator with household and calendar half-year fixed effects. The outcome is $\log(1 + y)$, where y is calories purchased (panel a) or expenditure (panel b) across all grocery categories. The omitted reference period is the half-year immediately before adoption. Event time is measured in half-years: period 0 is the half-year containing the household's first reported GLP-1 use, so households are treated for only part of it, and periods 1 to 3 are the three half-years that follow. The sample consists of GLP-1 adopter and never-adopter households observed from 2023H1 through 2025H2 (4,256 households; 23,078 household–half-year observations). Vertical bars denote 95 percent confidence intervals. Standard errors are clustered at the household level. *Source:* Authors' calculations using the NielsenIQ Consumer Panel.

In panel b, GLP-1 adoption reduces total grocery spending by 7.1% in the adoption half-year ($p < 0.001$). The decline persists over the next two half-years, at -4.3% ($p = 0.005$) and -4.5% ($p = 0.04$). Averaged over the post-adoption horizon, spending falls by 5.3% ($p < 0.001$). Our estimated spending effect is similar to that of Hristakeva et al. (2026), who report a 5.3% decline in grocery spending in the six months after adoption. Item counts in that study fall in proportion to spending, which the authors interpret as households buying less rather than trading down.

In contrast, we estimate a calorie effect that is not proportional to spending. Calories fall by −3.4% one half-year out and by an insignificant −1.5% ($p = 0.52$) two half-years out. Spending is at −4.3% and −4.5% over the same horizon ($p = 0.005$ and $p = 0.04$). A regression on the difference between the two outcomes also finds a gap in the adoption half-year, of 2.0 percentage points ($p = 0.04$), and of 1.9 percentage points pooled over the post-adoption horizon ($p = 0.08$). We interpret this gap cautiously, because the difference itself fails the joint test of parallel pre-adoption trends that spending and calories each pass ($p = 0.048$). In any case we find that the reduction in spending is greater than the reduction in calories. In the following subsections we explore the composition of the change in calories.

4.2 Nutrients

Figure 2 plots the event study for calories and eight nutrient aggregates, and Appendix Table A10 reports their average effects pooled over the four post-adoption half-years. The nutrient response follows the aggregate calorie response, peaking in the adoption half-year and fading thereafter. In the adoption half-year, we estimate reductions in six of the eight nutrients: total fat by 7.1%, dietary fiber by 6.8%, protein by 5.8%, sodium by 5.4%, saturated fat by 5.3%, and total carbohydrate by 3.6%. These declines persist one half-year out and shrink toward zero by the second. Pooled, dietary fiber (5.3%), protein, total fat, saturated fat, and sodium remain significant ($p < 0.01$). The decline is broad rather than targeted at any one nutrient, and it is largest for total fat.

Our results for total fat and saturated fat call for more cautious interpretation than the rest. Both carry a positive coefficient one year before adoption (+2.7% and +2.5%, both $p < 0.05$), indicating a pre-trend. No other nutrient has a lead distinguishable from zero at the 5 percent level. A downward drift into the reference period would overstate the reductions in fat.

Sugar and added sugar are the exceptions in both directions. Neither falls significantly at adoption, and two half-years out they are the only nutrients with positive coefficients significant at the 10 percent level (+5.2% and +7.2%).

Taken together, the results in Figure 2 suggest an initial change in the nutritional composition of the grocery basket away from fat, fiber, protein, and sodium, and towards sugar and carbohydrates. Over time, the reductions in fat, fiber, protein, and sodium attenuate, as we found for calories. At the same time, the sugar and carbohydrate contents increase, although we do not measure these changes precisely.

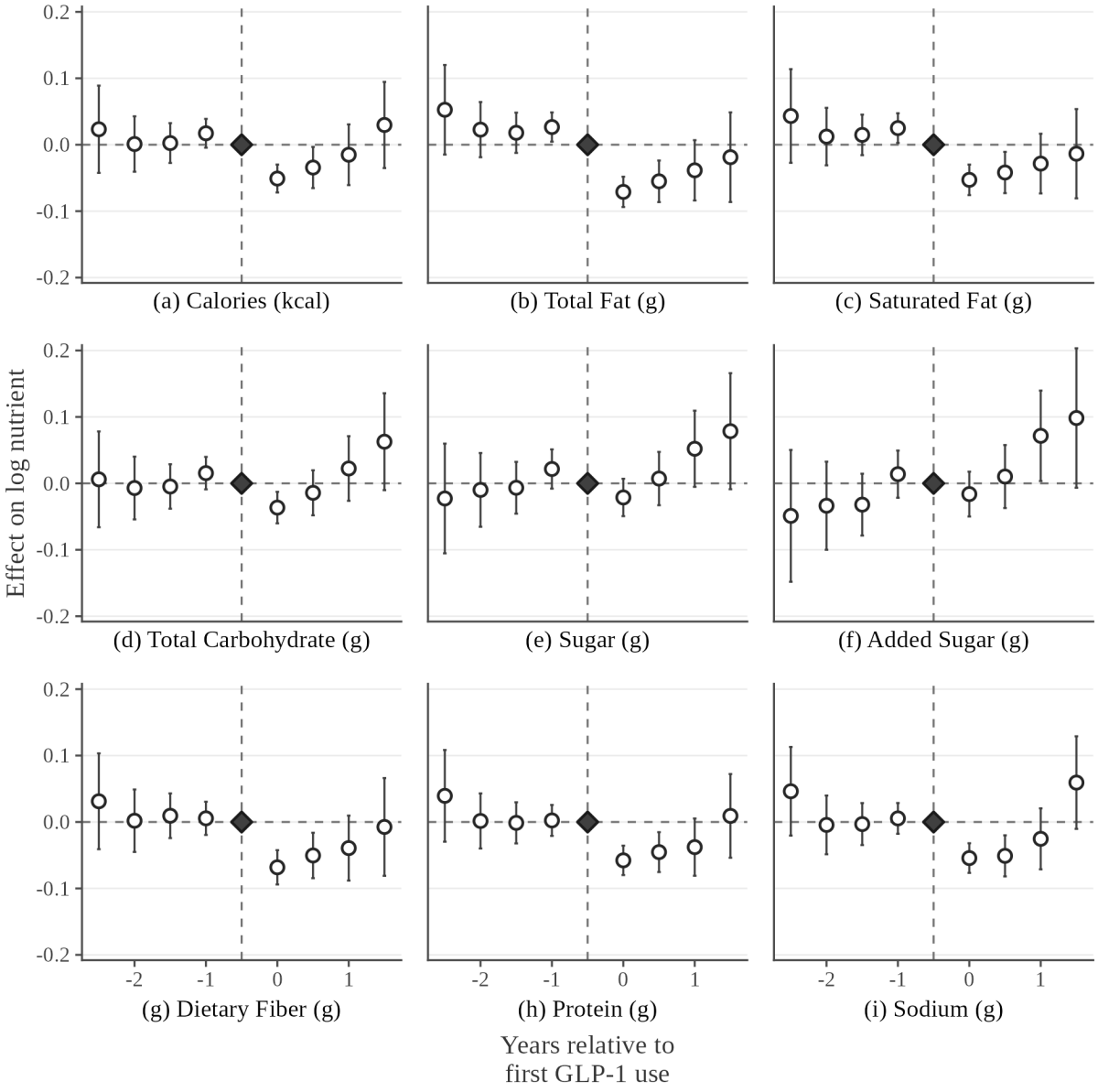


Figure 2: Effect of GLP-1 adoption on nutrients purchased, by event time

Notes: Each panel plots Sun and Abraham (2021) staggered difference-in-differences coefficients for log(1 + nutrient) aggregated across all grocery purchases, with household and calendar half-year fixed effects. The nine outcomes are the same ones reported in Appendix Table A10. Event time is measured in half-years and plotted in years: period 0 is the half-year containing the household's first reported GLP-1 use, so households are treated for only part of it. The diamond marks the omitted reference period, the half-year immediately before adoption. The sample consists of GLP-1 adopter and never-adopter households observed from 2023H1 through 2025H2 (4,256 households; 23,078 household–half-year observations). Vertical bars denote 95 percent confidence intervals. Standard errors are clustered at the household level. *Source:* Authors' calculations using the NielsenIQ Consumer Panel.

4.3 Effects by Food Group

Figure 3 plots the estimated treatment effects on calories and spending across all eleven food groups at each post-adoption period. Food groups are listed in ascending order by the adoption-half-year calorie coefficient.[2] In the adoption half-year, the reduction in total calories is driven by reductions across many food groups, with the biggest reductions in meat and beans (11.6%), salty snacks, and grains. Table fats and oils and vegetables have sizable point estimates that are not statistically significant at the 5 percent level, and prepared meals, beverages, and fruit barely change. A joint test rejects equal calorie effects across the eleven groups over all three post-adoption periods jointly ($p = 0.03$; Appendix Table A9).

Estimated treatment effects attenuate in later periods for most food groups, but the pattern is heterogeneous. For calories, the meat and beans estimates stay negative across all three periods. The cut deepens to 16.0% two half-years out, and meat and beans is the only group whose calorie cut stays significant in all three periods once we account for testing many food groups at once. The most durable calorie cut therefore lands on the basket's largest source of protein. In the first half-year after adoption, only calories from meat and beans and, at the 10 percent level, salty snacks are statistically significantly lower, and in the second half-year after adoption, only calories from meat and beans are statistically significantly lower. Confidence intervals for the caloric estimates are generally wider than those for the corresponding spending estimates, reflecting greater variability in caloric purchases.

Spending follows a similar ranking in the adoption half-year, with meat and beans down 8.2%. Two half-years out, meat and beans is still the deepest spending cut (-11.7%). In every period its calorie cut exceeds its spending cut, so spending per calorie in the group rises, consistent with households buying less of it while shifting toward products that cost more per calorie, such as leaner meats. Appendix E decomposes protein, sugar, fiber, and sodium by food group; meat and beans is the group whose cuts persist.

[2]These decompositions are exploratory. We correct for multiple testing only where a food-group effect persists to the second half-year after adoption. Elsewhere the reported p-values are unadjusted, and we read these panels as descriptive patterns that generate hypotheses rather than confirm them. Two corrections are reported. The q-values are Benjamini–Hochberg (Benjamini and Hochberg 1995) over all 66 coefficients in the figure (eleven groups × three periods × two outcomes). The Romano–Wolf adjusted p-values are stepdown (Romano and Wolf 2005) within a single outcome, so that family holds 33 coefficients. For meat and beans calories, the Romano–Wolf p-values are below 0.01 in the first two periods and 0.017 two half-years out, where the q-value is 0.006. For meat and beans spending two half-years out, the q-value is 0.012 and the Romano–Wolf p-value is 0.043. Appendix D gives the implementation.

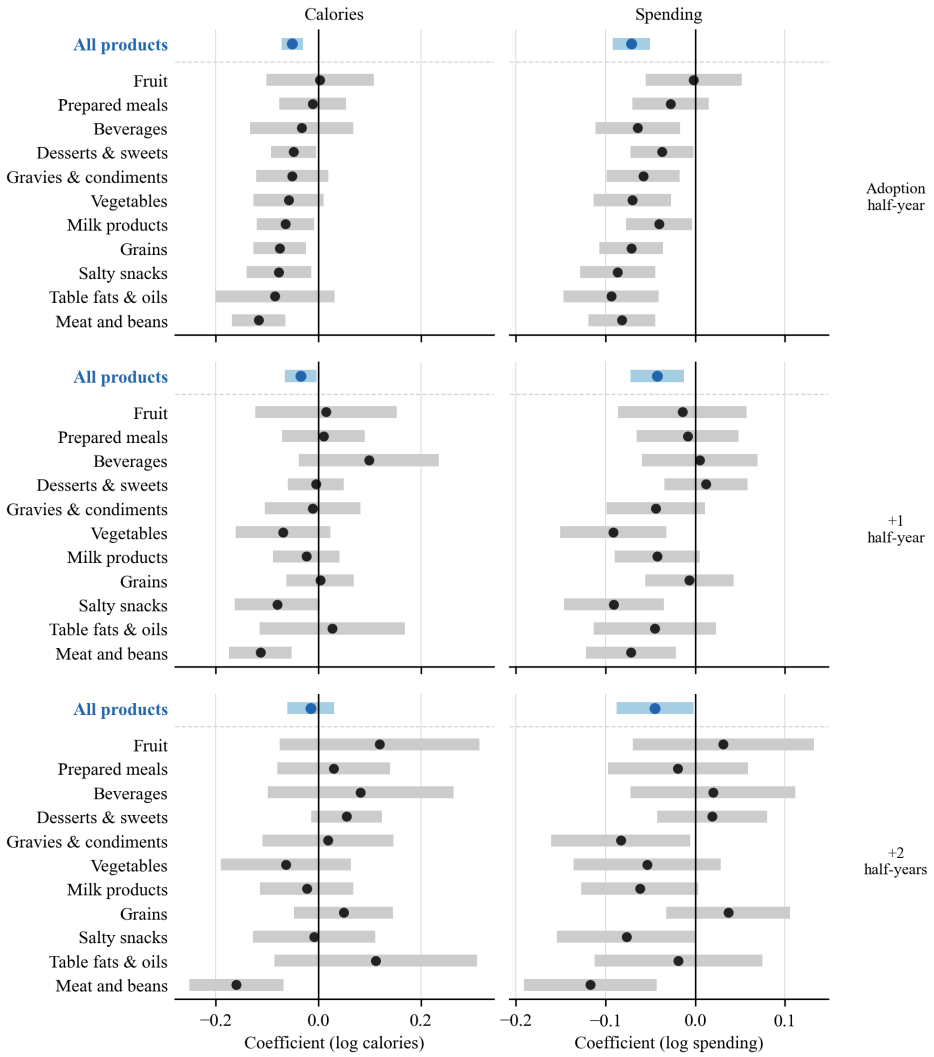


Figure 3: Effect of GLP-1 adoption on calories and grocery spending by food group, by post-adoption period

Notes: Each point shows the Sun and Abraham (2021) staggered difference-in-differences coefficient for log(1 + calories) (left column) and log(1 + expenditure) (right column) in the adoption half-year and the two half-years that follow (rows). The eleven food groups follow the USDA Economic Research Service (ERS) food-group classification (six of its seven Tier-1 groups, with "Other Foods" disaggregated into five Tier-2 subgroups), which defines groups by both nutritional content and food type (USDA-ERS 2016). The eleven ERS groups account for all but roughly 0.3 percent of grocery spending. The small residual is not attributed to a food group and carries no matched nutrient record, so the group-level decomposition need not sum exactly to the aggregate. Food groups are ordered by the adoption-half-year calorie coefficient (ascending). The highlighted "All products" row at the top of each panel reports the corresponding total-grocery (all-products) coefficient for the same outcome and period, as a benchmark. The sample consists of GLP-1 adopter and never-adopter households observed from 2023H1 through 2025H2, with household and calendar half-year fixed effects. Horizontal bars denote 95 percent confidence intervals. Standard errors are clustered at the household level. The estimation sample is the matched panel of 4,256 households (2,128 adopter–control pairs; 23,078 household–half-year observations); each food-group panel is filled to this complete grid, with half-years in which a household buys nothing in a group entered as zero spending, so every per-group regression uses the same sample. *Source:* Authors' calculations using the NielsenIQ Consumer Panel.

4.4 Heterogeneity by Drug Indication and Persistence of Use

We probe heterogeneity in the calorie and grocery spending responses along two dimensions of GLP-1 use: the labeled indication of the drug used (weight loss versus diabetes) and the persistence of use.

Figure 4 presents heterogeneous treatment effects for two household types defined by the labeled indication of the GLP-1 drug used: weight-loss–indicated and diabetes-indicated. Weight-loss–indicated households use at least one GLP-1 drug labeled for weight loss. Diabetes-indicated households use only GLP-1 drugs labeled for diabetes. These designations reflect the drugs' labeled indication, not an observed diagnosis or the household's stated motive for use.

Adopters of diabetes-indicated medications respond less than adopters of weight-loss–indicated medications. Averaged over the post-adoption horizon, calories purchased fall by 7.9% ($p < 0.001$) among adopters of weight-loss–indicated medications but by an insignificant 1.1% ($p = 0.49$) among adopters of diabetes-indicated medications. The two average calorie effects differ by 6.8 percentage points ($p = 0.009$). Dynamics also differ across indications. Among households using weight-loss–indicated medications, calories are 8.6% lower in the adoption half-year and still 8.6% ($p = 0.003$) lower one half-year out. Among diabetes-indicated households, calories fall by 3.1% ($p = 0.02$) in the adoption half-year and are back at baseline one half-year later. The pooled calorie decline is therefore driven almost entirely by households using weight-loss–indicated medications.

The two groups also differ in the breadth of their food-group response. Adopters of weight-loss–indicated medications cut calories across a broad set of staples and discretionary groups in the adoption half-year. Adopters of diabetes-indicated medications adjust a far narrower set. Among weight-loss–indicated households, calories fall significantly for table fats and oils (−31.2%), meat and beans (−18.6%), grains (−15.1%), salty snacks (−13.8%), milk products (−12.5%), and desserts and sweets (−10.8%). Among diabetes-indicated households, calories fall significantly only for meat and beans (−7.5%). The reductions in grains, salty snacks, and milk products estimated for the weight-loss–indicated households are absent or far weaker.

Over time, for weight-loss–indicated households, the calorie reduction in meat and beans deepens, while the effects fade for most other food groups. Two half-years out, meat and beans has the largest calorie reduction. Salty snacks stays large one half-year out (−21.0%, $p < 0.01$) before fading, while the grains reduction fades after the adoption half-year, as does that for table fats and oils. For diabetes-indicated households the response is weaker and more mixed. Meat and beans stays negative but imprecise (−7.1%, $p = 0.18$ two half-years out). Desserts and sweets turn positive after adoption and reach +10.4% ($p = 0.01$) two half-years out.

Figure 5 splits adopters by whether they report discontinuing GLP-1 use in a later survey wave.

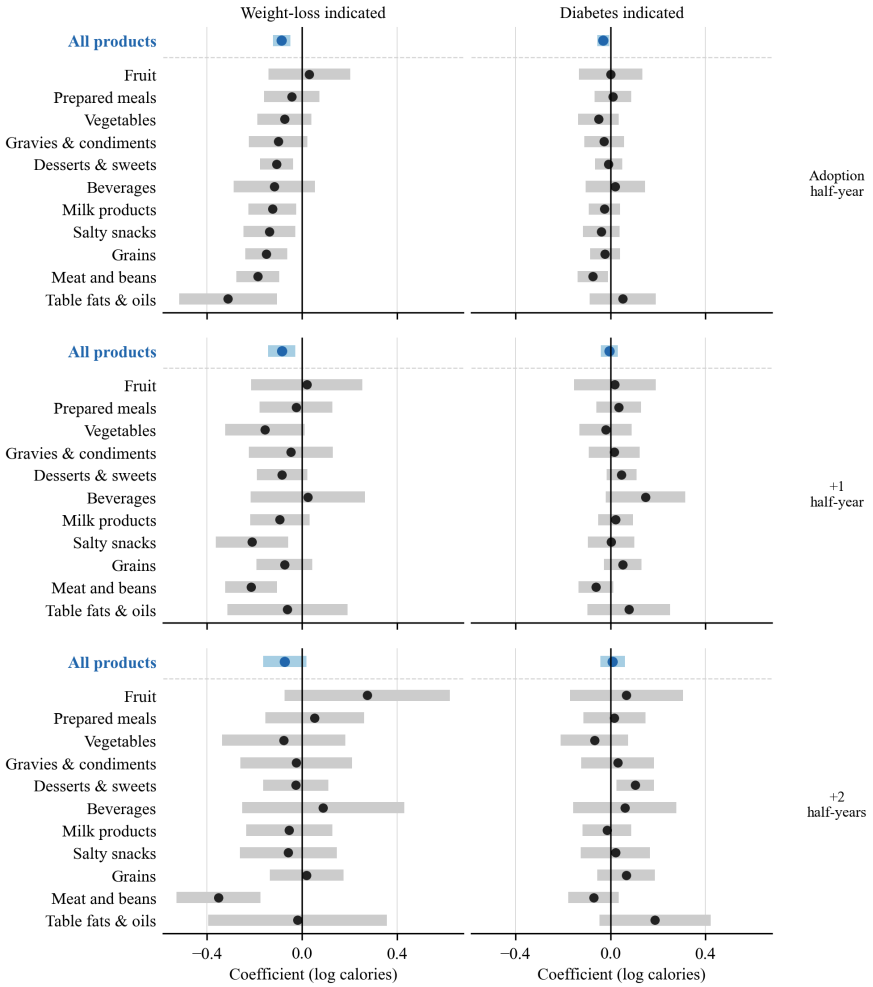


Figure 4: Effect of GLP-1 adoption on calories purchased by drug indication and food group, by post-adoption period

Notes: Each point shows the Sun and Abraham (2021) staggered difference-in-differences coefficient for log(1 + calories) for weight-loss–indicated (left column) and diabetes-indicated (right column) households in the adoption half-year and the two half-years that follow (rows). Weight-loss–indicated households use at least one GLP-1 drug labeled for weight loss. Diabetes-indicated households use only GLP-1 drugs labeled for diabetes. Food groups are ordered by the adoption-half-year weight-loss–indicated calorie coefficient (ascending). The highlighted "All products" row at the top of each panel reports the corresponding total-calories (all-products) coefficient for that same indication group and period, as a benchmark. The sample consists of GLP-1 adopter and never-adopter households observed from 2023H1 through 2025H2, with household and calendar half-year fixed effects. Horizontal bars denote 95 percent confidence intervals. Standard errors are clustered at the household level. Estimation samples comprise 1,662 households in the weight-loss–indicated subsample (8,982 household–half-year observations) and 2,594 in the diabetes-indicated subsample (14,096 observations). Each subsample consists of the adopters carrying that indication together with their one-to-one matched never-adopter controls, so both figures count households rather than adopters, and the controls carry no indication of their own; adopters whose drug indication could not be determined are excluded, along with their matched partners, from both subsamples. Each food-group panel is filled to its subsample's complete grid, with half-years of no purchase in a group entered as zero, so per-group regressions use the same sample. *Source:* Authors' calculations using the NielsenIQ Consumer Panel.

About one in five adopters report stopping at some point. Discontinuation is not random, and we do not know why households stop. We do not claim to identify the effect of discontinuing GLP-1 medications.

Persistent users cut calories broadly at adoption, and the cuts in meat and beans and vegetables grow larger over time. In the adoption half-year, calories fall significantly for table fats and oils (−13.4%), salty snacks (−11.5%), meat and beans (−11.0%), and grains (−8.5%). Meat and beans deepens to −21.4% two half-years out ($p < 0.01$), and vegetables from −6.8% to −14.3% ($p = 0.049$). Total calories among persistent users fall in the adoption half-year (−5.6%, $p < 0.01$) and are no longer significant two half-years out (−2.7%, $p = 0.28$), even as those two groups keep falling. Their aggregate spending instead stays lower, at −7.6% in the adoption half-year and −7.1% two half-years out (both $p < 0.01$).

Discontinuing households cut calories significantly in only one food group in the adoption half-year: meat and beans (−13.9%, $p < 0.05$). That reduction dissipates once households discontinue. Total calories in this subsample are never distinguishable from zero (−2.2% at adoption, $p = 0.45$). The aggregate spending effect is −4.4% in the adoption half-year ($p = 0.10$) and indistinguishable from zero thereafter. Several food groups even drift positive two half-years out, including beverages (+28.6%), table fats and oils (+31.2%), and salty snacks (+20.1%), none significant. The pattern is consistent with a rebound but is not evidence of one. The contrast reinforces that the sustained response tracks continued GLP-1 exposure, not a one-time adjustment at adoption. The smaller, less precise discontinuation subsample makes this split suggestive rather than dispositive.

4.5 Robustness

The nutrient outcomes rest on a match between UPCs and nutrition records, so we check that incomplete coverage does not generate the estimated declines. In the estimation sample, coverage of the UPC–nutrient match is at least 94.8 percent of spending for every nutrient in the aggregate and is balanced across adopters and matched controls (Appendix Table A3). Re-estimating on only the directly matched nutrient values yields declines of essentially the same size (Appendix Table A4), so incomplete coverage does not generate the nutrient response.

We re-estimate the baseline specification using several alternative event-study estimators to check whether our results are sensitive to the choice of staggered difference-in-differences method. These include the standard TWFE estimator, the Callaway and Sant'Anna (2021) estimator, the Borusyak et al. (2024) imputation estimator, and the Roth and Sant'Anna (2023) efficient estimator. Appendix Figure A1 presents the event-study coefficients for total calories purchased across these different methods. Pre-adoption coefficients are flat under most estimators, and all five show a decline in

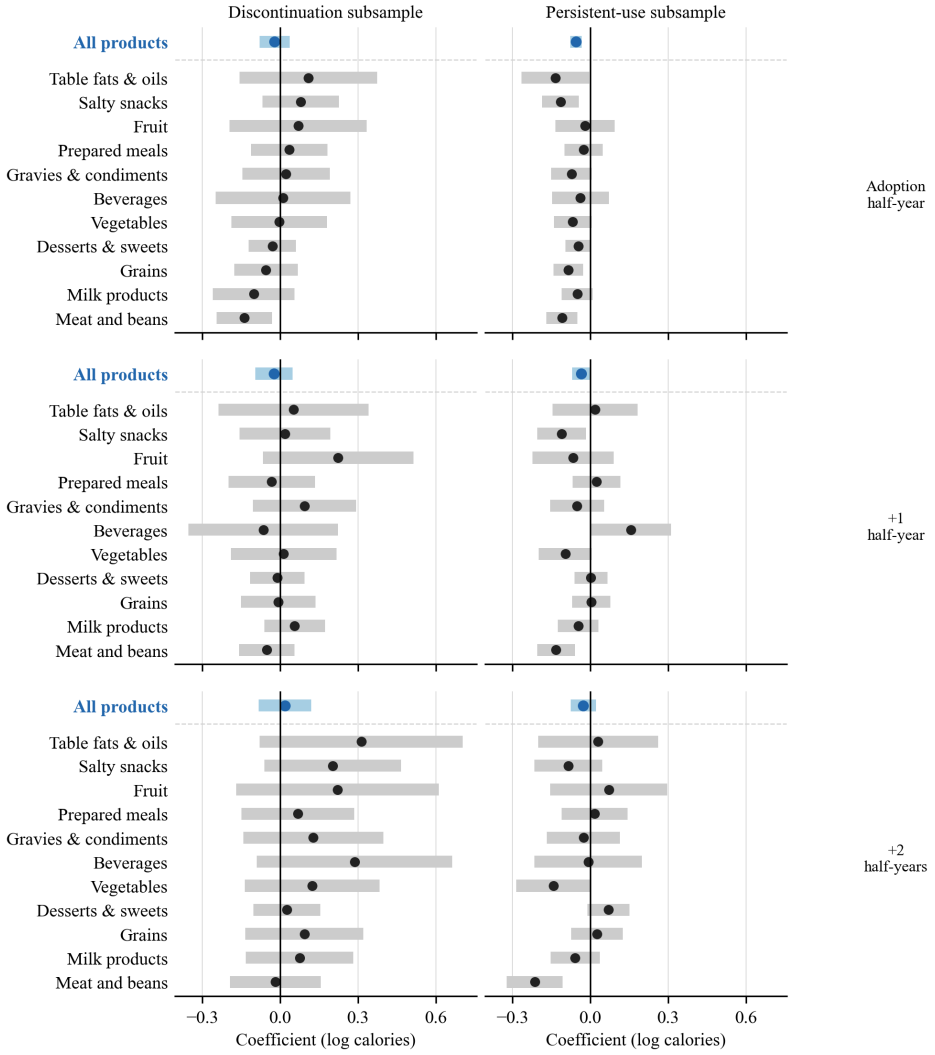


Figure 5: Effect of GLP-1 adoption on calories purchased by persistence of use and food group, by post-adoption period

Notes: Each point shows the Sun and Abraham (2021) staggered difference-in-differences coefficient for log(1 + calories) for the discontinuation subsample (left column, households that eventually stopped GLP-1 use) and the persistent-use subsample (right column) in the adoption half-year and the two half-years that follow (rows). Each subsample is matched to its own set of 1:1 never-treated control households. Food groups are ordered by the adoption-half-year discontinuation-subsample calorie coefficient (ascending). The highlighted "All products" row at the top of each panel reports the corresponding total-calories (all-products) coefficient for that same subsample and period, as a benchmark. The sample consists of GLP-1 adopter and never-adopter households observed from 2023H1 through 2025H2, with household and calendar half-year fixed effects. Horizontal bars denote 95 percent confidence intervals. Standard errors are clustered at the household level. Estimation samples comprise 816 households in the discontinuation subsample (4,413 household–half-year observations) and 3,440 in the persistent-use subsample (18,665 observations). Both counts include each adopter's matched never-adopter partner, so they count households rather than adopters. Each food-group panel is filled to its subsample's complete grid, with half-years of no purchase in a group entered as zero, so per-group regressions use the same sample. *Source:* Authors' calculations using the NielsenIQ Consumer Panel.

the adoption half-year. The efficient estimator finds a smaller decline in the adoption half-year and none afterward. Shifting the reference period back to $e = -2$, one year before adoption, shows no anticipation at the half-year frequency (Appendix Figure A2).

Pre-adoption spending coefficients in Figure 1 cluster near zero, supporting parallel trends. For calories, the $e = -3$ and $e = -2$ leads are both statistically insignificant ($p = 0.87$ and $p = 0.12$). A joint Wald test of the pre-adoption leads does not reject parallel trends for either series (spending $p = 0.69$; calories $p = 0.28$).

Flatness before adoption is necessary but not sufficient, so we apply the Rambachan and Roth (2023) honest difference-in-differences bounds (Appendix Figure A3). They give the largest post- to pre-adoption violation ratio $\bar{M}$ at which an effect stays distinguishable from zero, and so bound how large a confounder timed to adoption could be. The adoption-half-year spending effect is robust, surviving to $\bar{M} = 1.6$, while calories break down at $\bar{M} = 0.7$. The pooled effects are less robust (pooled spending $\bar{M} = 0.6$, pooled fiber $\bar{M} = 0.3$). The adoption-half-year fiber (6.8%, $\bar{M} = 0.8$) and protein (5.8%, $\bar{M} = 0.6$) declines are sturdier. The later horizons are fragile for a second reason: they rest on progressively fewer adoption cohorts, with the second half-year after adoption drawing on two cohorts totaling 880 households (Appendix Table A8). We therefore rest the causal interpretation on the adoption half-year, and report coefficients period by period rather than the pooled effect alone.

A remaining concern is that adoption is bundled with a decision to diet. Adoption of a weight-loss–labeled drug is the case most likely to be bundled with dieting intent, and, as Figure 4 shows, it is also the subsample with the larger aggregate calorie and spending responses. But weight-loss formulations are typically dosed higher, so the gap is equally consistent with a stronger pharmacological effect. The split cannot separate the two.

Throughout, we read our estimates as the effect of the adoption episode, the medication plus whatever behavior change accompanies starting it, not of the drug alone. Separating the two is not necessary, because in practice households do not take the medication without it. If the decline came only from households that had already decided to diet, it should disappear among adopters of diabetes-indicated medications, whose adoption is least plausibly driven by a wish to lose weight. It does not: their calories purchased still fall in the adoption half-year (Figure 4). This comparison cannot rule out a diet change that begins exactly when the medication does. Such a change would leave the pre-adoption coefficients flat, so neither the pre-trend tests nor the honest bounds can detect it.

5 Conclusion

This paper uses detailed data on grocery transactions to show how GLP-1 adoption changes the caloric and nutrient content of household grocery purchases. By linking UPC-level grocery transactions to product-level nutrition information, we observe dimensions of the household purchasing response that cannot be recovered from grocery expenditures, item counts, or broad food categories alone. A decline in food spending does not reveal how many calories households give up, and a decline in calories does not reveal which nutrients they cut.

We find that adoption of GLP-1 medications reduces the calories in household grocery purchases by 5.1% in the adoption half-year. The reduction is smaller one half-year later and no longer distinguishable from zero two half-years out. Pre-adoption trends are flat in the main specification, and most alternative estimators give a similar adoption-half-year decline. Most nutrients fall with calories: fiber, protein, sodium, total fat, and saturated fat all decline in the adoption half-year and remain lower when the post-adoption periods are pooled. The exceptions are sugar and added sugar, which never fall significantly and rise two half-years after adoption (+5.2% and +7.2%, both $p < 0.10$). The grocery basket of GLP-1 adopters therefore shrinks without shrinking proportionally: its composition shifts away from protein and toward sugar.

The food-group results help explain this pattern. The largest, most persistent calorie reduction is in meat and beans, the largest source of protein in the grocery basket. Calories from meat and beans remain significantly lower two half-years after adoption. By contrast, for several other food groups we estimate caloric reductions initially, but the effects attenuate over time. Which foods households stop purchasing matters for the resulting changes in calories and nutrients.

These results have implications for understanding the nutritional consequences of GLP-1 adoption. In our data, nutrients associated with meat and beans, including protein, decline following adoption, while sugar does not. Thus, the nutritional consequences of GLP-1 adoption cannot be inferred from changes in grocery expenditures or calories alone; examining the nutrient content of purchases provides a different and more detailed picture of how food purchasing changes.

This point is particularly relevant given clinical concerns about maintaining adequate nutrition during GLP-1 treatment. Appetite suppression changes how much food consumers want, but it does not by itself determine which foods and nutrients they reduce. The results suggest that changes in household food purchases accompanying GLP-1 treatment may have nutritional consequences that require attention beyond the overall reduction in food purchases. Dietary guidance may therefore play an important role alongside medication use, particularly if consumers reduce purchases of foods that provide nutrients of concern during weight loss.

The expenditure results provide complementary evidence on the broader economic response. GLP-

1 adoption reduces grocery spending by 7.1% in the adoption half-year, consistent with the emerging transaction-based literature on the effects of these medications on household food markets. But expenditures alone cannot characterize the nutritional consequences of this response. Observing expenditures, calories, and nutrients in the same purchases makes it possible to distinguish changes in the economic value of the grocery basket from changes in its caloric and nutritional content.

The response is also larger for adopters of weight-loss–indicated medications and for households that sustain use. These patterns are consistent with a response that grows with dose and continued use, although our analysis cannot distinguish a stronger pharmacological effect from other explanations, including a stronger motivation to lose weight. At the same time, our estimates should be interpreted as the effects of the adoption episode rather than the pharmacological effect of the medication alone. The adoption episode is the medication plus whatever behavior change accompanies starting it.

Several limitations qualify the interpretation of the results. GLP-1 use, medication indication, and discontinuation are self-reported in the GLP-1 survey rather than drawn from pharmacy or insurance-claims records. The survey is fielded semiannually, which limits the precision with which adoption and discontinuation can be timed. In addition, the panel records food purchased by households, not food consumed. We also do not observe food eaten away from home, plate waste, or changes in household food inventories. The nutrient measures therefore describe the calories and nutrients contained in household grocery purchases rather than actual dietary intake.

A further limitation is that the purchases we observe are equilibrium outcomes in food markets, rather than measures of consumer demand alone. Food manufacturers and retailers may respond to the growing population of GLP-1 users by changing product assortment, introducing or reformulating products, adjusting prices and promotions, or changing marketing strategies (e.g., Cengiz and Rojas 2024). Because matched never-adopters shop in the same markets over the same half-years, supply-side changes common to all households are differenced out, but responses targeted at GLP-1 users are not.

Several extensions follow naturally. The first is linking purchases more closely to consumption. Combining grocery transactions with information on food away from home, household inventories, food waste, or dietary intake would provide a more complete picture of how GLP-1 adoption changes nutrition. A second is to test directly whether the nutrient density of the basket changes, as our estimates for sugar suggest. Such analysis would help distinguish reductions in overall food purchases from changes in the mix of nutrients households purchase. A third direction is to study how food manufacturers and retailers respond to the diffusion of GLP-1 medications. Studying these responses would help distinguish changes generated directly by consumers from changes generated by firms adapting to a new and growing segment of the food market.

More broadly, the results raise questions about the mechanisms and welfare consequences of this change in food markets. GLP-1 medications may affect purchases through reduced appetite, changes in food preferences, changes in household routines, or deliberate efforts to modify diet following adoption. Distinguishing among these consumer-side mechanisms and supply-side responses would help explain why some food groups and nutrients respond more strongly than others and would provide a basis for evaluating the welfare consequences of GLP-1 diffusion for consumers and firms throughout the food supply chain.

The rapid diffusion of GLP-1 medications has created a potentially important new source of change in food markets. Our results show that this change extends beyond grocery expenditures. GLP-1 adoption changes the calories and nutrients contained in household grocery purchases, and those changes depend on which foods households reduce. Measuring these nutritional dimensions directly is therefore important for understanding how the growing use of GLP-1 medications changes household food purchases. A fuller understanding of the consequences for consumers and the food system will ultimately require studying both sides of the market: how consumers change their food purchasing behavior and how food manufacturers and retailers respond.

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A Data and Measurement

A.1 Nutrient Coverage

Calories and each nutrient are constructed at the item level and summed to the household–half-year. For every item in a household's basket, we multiply its per-unit nutrient content, taken from the UPC–nutrient match, by the number of units purchased, and add across items. The following four tables document the spend-weighted nutrient coverage of that match in the NielsenIQ purchase data used in the analysis.

Coverage exceeds 95 percent for nearly every ERS food group and nutrient (Table A1). The two exceptions are dietary fiber within beverages (about 88 percent) and added sugar within fruit (about 94 percent). Neither is material. Beverages contain almost no fiber, so unmatched drink items contribute negligibly to the fiber totals regardless. Added sugar is one of the nutrient aggregates we estimate, but its coverage across all food groups is 98.3 percent (Table A2), so the shortfall within fruit changes the totals little.

Table A2 decomposes overall coverage by data provenance, distinguishing direct label matches, logic imputation, and proxy (donor) fill from unfilled transactions. Within the matched estimation sample, coverage is balanced across adopters and matched controls and does not drift differentially after adoption (Table A3).

Table A4 re-estimates the treatment effects on calories and the eight nutrient aggregates under two measurement variants. The first uses only directly matched nutrient values. The second rescales each household–half-year nutrient total by its matched share of spending, which removes products without a nutrient record from both the numerator and the spending-share denominator. The two variants probe the proxy imputation and incomplete nutrient coverage, respectively.

A.2 Expenditure per Calorie by Food Group

Table A5 reports expenditure per calorie (dollars per 1,000 kcal) for GLP-1 user households and matched controls across the pre-initiation and initiation periods.

Upon first GLP-1 use, expenditure per calorie declines in seven of the eleven food groups relative to the pre-initiation period, with the largest proportional drops in table fats and oils, fruit, and salty snacks. It rises in meat and beans, desserts and sweets, milk products, and gravies and condiments. User households pay more per calorie than controls in most food groups in both periods, though the control column averages over all half-years in the estimation window rather than at a single event time, so that gap is descriptive rather than a like-for-like comparison.

Table A1: Spend-Weighted Nutrient Match Rate by ERS Food Group (%)

	Calories	**Protein**	**Fat**	**Sat. fat**	**Sugar**	**Add. sugar**	**Sodium**	**Fiber**	**Carb.**
Grains	99.6	99.6	99.6	99.6	99.6	99.5	99.6	99.5	99.6
Vegetables	99.5	99.2	99.4	99.5	98.7	95.0	99.4	99.3	99.1
Fruit	96.3	97.2	97.4	97.7	97.1	93.6	97.4	97.3	97.4
Milk products	99.7	99.6	99.7	99.7	99.6	97.3	99.6	95.7	99.6
Meat and beans	99.5	99.5	99.5	99.5	99.5	98.6	99.5	98.7	99.4
Prepared meals	99.5	99.4	99.4	99.4	99.4	99.3	99.5	99.0	99.4
Table fats & oils	99.8	99.8	99.8	99.8	99.8	99.6	99.8	99.8	99.8
Gravies & condiments	99.6	99.6	99.6	99.6	99.6	99.2	99.6	99.4	99.6
Beverages	99.7	96.5	99.6	99.6	99.7	99.5	99.7	87.7	99.6
Desserts & sweets	99.0	98.9	99.0	99.0	99.0	98.8	99.0	98.0	99.0
Salty snacks	99.4	99.2	99.4	99.4	99.4	98.7	99.4	98.2	99.4
Not in ERS	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
All	**99.1**	**98.4**	**99.1**	**99.2**	**99.1**	**98.3**	**99.2**	**95.6**	**99.1**

Notes: All match rates are spend-weighted: each cell equals the share of dollars spent in that food group for which a nutrient value is available (direct label match or imputed). Direct label matches use NielsenIQ product attributes and Syndigo panel nutrient data, together with the USDA FoodData Central reference composition used for fruit and vegetables and the package-weight density fill for goods packaged by count. Imputed values come from a three-tier nearest-neighbor proxy: tier 1 uses the within-brand subcategory mean, tier 2 uses the outside-brand subcategory mean, and tier 3 uses the full subcategory average. Food groups follow the USDA Economic Research Service (ERS) food-group classification (USDA-ERS 2016). Products that cannot be mapped to an ERS food group are shown in the "Not in ERS" row; this category accounts for approximately 0.3% of total spend and has no nutrient library coverage. The final (All) row is the spend-weighted match rate over all major-brand grocery spending. Purchases whose UPC has no nutrient record are counted as unmatched within their food group (or in the Not in ERS row when unclassifiable); they are not dropped from the denominator. *Source:* Authors' calculations using the NielsenIQ Consumer Panel with NielsenIQ, Syndigo, and USDA FoodData Central nutrient data.

Table A2: Spend-Weighted Nutrient Coverage by Data Provenance (%, Full Sample)

	Direct match	**Logic imputation**	**Proxy fill**	**Covered total**	**No nutrient data**
Calories	79.3	0.0	19.8	**99.1**	0.9
Protein	81.2	0.1	17.1	**98.4**	1.7
Total fat	82.0	0.0	17.1	**99.1**	0.9
Saturated fat	77.8	4.2	17.1	**99.2**	0.8
Sugar	59.7	6.9	32.4	**99.1**	0.9
Added sugar	43.2	12.5	42.5	**98.3**	1.7
Sodium	82.0	0.1	17.1	**99.2**	0.9
Dietary fiber	77.0	1.4	17.1	**95.6**	4.4
Total carbohydrate	82.0	0.1	17.1	**99.1**	0.9

Notes: Each row reports the spend-weighted share of all major-brand grocery spending (approximately 90.4 million transactions) attributable to each data-provenance bucket; the row entries sum to 100%, so the Covered total and No nutrient data columns add to 100%, up to rounding of the displayed digits. Direct match combines original NielsenIQ label-level nutrient data with the Syndigo backfill, the USDA FoodData Central reference composition for fruit and vegetables, and imputation derived from package weight. Logic imputation applies zero-fill rules when a macro-nutrient value implies another is zero (for example, zero calories implies zero protein). Proxy fill uses a three-tier donor-mean imputation (within-brand, outside-brand, and subcategory averages). Covered total sums the direct, logic, and proxy columns and equals the All row of Table A1 (within 0.5 pp). No nutrient data is the share of spending for which no nutrient value was obtained from any source. It combines two cases: UPCs absent from the nutrient master table (approximately 0.7 pp, the same products for every nutrient—predominantly food and beverage items not covered by the nutrient sources, such as energy drinks, flavored sodas, and cookies), and UPCs present in the master table for which no subcategory donor was available for imputation (ranging from approximately 0.1 pp for saturated fat to 3.7 pp for dietary fiber). *Source:* Authors' calculations using the NielsenIQ Consumer Panel with NielsenIQ, Syndigo, and USDA FoodData Central nutrient data.

Table A3: Spend-Weighted Nutrient Coverage by Treatment Status and Adoption Period (%)

	Adopters			Matched controls			
	Pre	**Post**	**Δ (pp)**	**Pre**	**Post**	**Δ (pp)**	**Diff.-in-diff. (pp)**
Calories	99.0	99.6	+0.58	98.9	99.6	+0.68	−0.10
Protein	97.5	98.9	+1.42	98.3	98.4	+0.11	+1.31
Total fat	99.0	99.6	+0.58	98.9	99.6	+0.69	−0.11
Saturated fat	99.0	99.6	+0.59	98.9	99.6	+0.69	−0.10
Sugar	99.0	99.5	+0.58	98.9	99.6	+0.70	−0.12
Added sugar	98.1	98.7	+0.61	98.1	98.8	+0.69	−0.08
Sodium	99.0	99.6	+0.59	98.9	99.6	+0.70	−0.11
Dietary fiber	94.8	96.5	+1.68	95.4	95.4	+0.03	+1.65
Total carbohydrate	99.0	99.6	+0.57	98.9	99.6	+0.70	−0.13

Notes: Each cell reports the spend-weighted nutrient coverage rate—the share of dollars spent for which a nutrient value is available (direct label match or imputed)—within the matched estimation sample (4,256 households observed from 2023H1 through 2025H2), split by treatment status and adoption period. Pre (Post) pools transactions in calendar half-years before (from) the household's first GLP-1 use; matched control households inherit the adoption half-year of their matched adopter, so the pre/post split is defined symmetrically in both groups. Δ is the within-group post-minus-pre change in coverage, in percentage points; the difference-in-differences column is the adopter Δ minus the control Δ. A negative difference-in-differences entry would indicate that adopters differentially drift toward products without nutrient records after adoption, which would mechanically depress measured nutrient purchases; the most negative entry is −0.13 percentage points (total carbohydrate), too small to generate the estimated declines. Coverage definitions and imputation tiers follow Table A2. *Source:* Authors' calculations using the NielsenIQ Consumer Panel with NielsenIQ, Syndigo, and USDA FoodData Central nutrient data.

Table A4: Nutrient ATT Robustness: Direct-Match-Only and Coverage-Adjusted Aggregates

	ATT	SE	p-value
Calories	−0.034***	(0.012)	0.006
direct match only	−0.033**	(0.013)	0.011
coverage adjusted	−0.034***	(0.012)	0.006
Protein	−0.046***	(0.012)	0.000
direct match only	−0.047***	(0.013)	0.000
coverage adjusted	−0.047***	(0.012)	0.000
Dietary fiber	−0.053***	(0.014)	0.000
direct match only	−0.054***	(0.014)	0.000
coverage adjusted	−0.055***	(0.014)	0.000
Sugar	0.008	(0.016)	0.637
direct match only	0.022	(0.019)	0.265
coverage adjusted	0.008	(0.016)	0.638
Sodium	−0.040***	(0.013)	0.002
direct match only	−0.038***	(0.013)	0.004
coverage adjusted	−0.040***	(0.013)	0.002
Total fat	−0.057***	(0.013)	0.000
direct match only	−0.056***	(0.013)	0.000
coverage adjusted	−0.057***	(0.013)	0.000
Saturated fat	−0.042***	(0.013)	0.001
direct match only	−0.042***	(0.013)	0.001
coverage adjusted	−0.042***	(0.013)	0.001
Total carbohydrate	−0.012	(0.014)	0.374
direct match only	−0.011	(0.014)	0.425
coverage adjusted	−0.012	(0.014)	0.381
Added sugar	0.016	(0.019)	0.414
direct match only	0.042*	(0.025)	0.092
coverage adjusted	0.015	(0.019)	0.428

Notes: Each row reports the Sun and Abraham (2021) staggered difference-in-differences average treatment effect on $\log(1 + \text{quantity purchased})$ of the indicated nutrient, with household and calendar half-year fixed effects and standard errors clustered at the household level (identical specification and sample as the baseline nutrient results: 4,256 households; 23,078 household–half-year observations spanning 2023H1 through 2025H2). Baseline rows use the full nutrient aggregates, in which missing label values are proxy-imputed (donor means; see Table A2). Direct-match rows re-aggregate the same transactions using only UPCs whose nutrient value comes from a direct label match (77–90% of purchased quantities). Coverage-adjusted rows rescale each household–half-year nutrient total by the household's matched share of spending in that half-year, excluding UPCs without a nutrient record from both the numerator and the spending-share denominator, so that differential drift in nutrient coverage cannot mechanically generate a decline. Direct-match estimates stay within 0.005 of the baseline for seven of the nine nutrients (the exceptions are sugar and added sugar), and coverage-adjusted estimates for all nine. No nutrient changes significance at the 5% level under either alternative. Direct-match standard errors are equal to or larger than the baseline, as expected when the filter discards part of the sample. Imputation and incomplete coverage therefore do not generate the estimated nutrient declines. *, **, and *** denote significance at the 10%, 5%, and 1% levels. *Source:* Authors' calculations using the NielsenIQ Consumer Panel with NielsenIQ, Syndigo, and USDA FoodData Central nutrient data.

Table A5: Half-year average expenditure per calorie by food group

Category	GLP-1 User Households		Controls
	Pre (6 months)	Initiation (6 months)	
	(USD per 1,000 kcal)		
Number of households	2,128	2,128	2,128
All Food	5.07	5.02	4.71
Beverages	5.57	5.47	4.63
Desserts, sweets, and candies	4.31	4.47	3.98
Fruit	10.10	9.60	9.42
Grains	3.33	3.21	3.07
Gravies, sauces, condiments, and spices	6.47	6.49	6.49
Meat and beans	6.60	6.72	6.56
Milk products	4.95	5.00	4.85
Prepared meals, sides, and salads	5.36	5.33	5.38
Salty snacks	3.38	3.24	3.22
Table fats, oils, and salad dressings	2.21	1.94	1.97
Vegetables	13.37	13.11	12.68

Notes: Each cell reports weighted-average expenditure per calorie (USD per 1,000 kcal) computed as total real expenditure divided by total purchased calories, scaled to a 1,000 kcal basis. This table shares the kilocalorie unit of the purchased-calories table, so a decline means expenditure per calorie falls. Because it is a ratio of two basket aggregates, such a decline reflects recomposition of the basket and not necessarily any change in the prices of individual products. The sample is the estimation sample, the same sample as the event-study regressions; a household enters a column whenever it is observed in the corresponding period. Column 1: same user households in the half-year before first GLP-1 use (pre-treatment); column 2: 2024H1–2025H2 GLP-1 user households in the half-year in which their first GLP-1 use falls (initiation half-year); column 3: 1:1 matched controls, averaged over all half-years in the estimation window rather than at a single event time; the within-adopter change across columns 1–2, not the gap with column 3, is the comparison of interest. "Number of households" reports the count of distinct households in the estimation sample for each group. The household count is the number of distinct households in that estimation sample. *Source:* Authors' calculations using the NielsenIQ Consumer Panel.

A.3 Food-Group Shares of Protein and Sugar Purchases

Table A6: Share of half-year protein purchases by food group

Category	GLP-1 User Households		
	Pre (6 months)	Initiation (6 months)	Controls
	(percent of half-year protein purchases)		
Number of households	2,128	2,128	2,128
All Food	100.00%	100.00%	100.00%
Beverages	4.40%	4.53%	3.51%
Desserts, sweets, and candies	7.18%	7.04%	7.10%
Fruit	0.59%	0.65%	0.66%
Grains	12.01%	12.12%	12.87%
Gravies, sauces, condiments, and spices	1.53%	1.52%	1.64%
Meat and beans	25.99%	25.37%	26.30%
Milk products	21.16%	21.58%	22.29%
Prepared meals, sides, and salads	16.62%	16.03%	15.20%
Salty snacks	6.90%	7.48%	6.59%
Table fats, oils, and salad dressings	0.08%	0.08%	0.08%
Vegetables	3.53%	3.59%	3.76%

Notes: Each cell reports the food group's share of total protein purchases in the column period (percent). Group-level means of half-year protein per adult equivalent are computed exactly as in the expenditure table, then normalized by the sum across the eleven food groups, so each column sums to 100. The sample is the estimation sample, the same sample as the event-study regressions; a household enters a column whenever it is observed in the corresponding period. Column 1: same user households in the half-year before first GLP-1 use (pre-treatment); column 2: 2024H1–2025H2 GLP-1 user households in the half-year in which their first GLP-1 use falls (initiation half-year); column 3: 1:1 matched controls, averaged over all half-years in the estimation window rather than at a single event time; the within-adopter change across columns 1–2, not the gap with column 3, is the comparison of interest. "Number of households" reports the count of distinct households in the estimation sample for each group. The household count is the number of distinct households in that estimation sample. *Source:* Authors' calculations using the NielsenIQ Consumer Panel.

Table A7: Share of half-year sugar purchases by food group

Category	GLP-1 User Households		
	Pre (6 months)	Initiation (6 months)	Controls
	(percent of half-year sugar purchases)		
Number of households	2,128	2,128	2,128
All Food	100.00%	100.00%	100.00%
Beverages	58.34%	58.01%	60.98%
Desserts, sweets, and candies	19.10%	19.28%	17.78%
Fruit	3.91%	4.08%	4.03%
Grains	3.46%	3.46%	3.37%
Gravies, sauces, condiments, and spices	2.26%	2.16%	2.07%
Meat and beans	0.99%	0.95%	0.96%
Milk products	6.24%	6.29%	5.88%
Prepared meals, sides, and salads	2.69%	2.72%	2.18%
Salty snacks	1.49%	1.58%	1.32%
Table fats, oils, and salad dressings	0.29%	0.27%	0.26%
Vegetables	1.23%	1.22%	1.19%

Notes: Each cell reports the food group's share of total sugar purchases in the column period (percent). Group-level means of half-year sugar per adult equivalent are computed exactly as in the expenditure table, then normalized by the sum across the eleven food groups, so each column sums to 100. The sample is the estimation sample, the same sample as the event-study regressions; a household enters a column whenever it is observed in the corresponding period. Column 1: same user households in the half-year before first GLP-1 use (pre-treatment); column 2: 2024H1–2025H2 GLP-1 user households in the half-year in which their first GLP-1 use falls (initiation half-year); column 3: 1:1 matched controls, averaged over all half-years in the estimation window rather than at a single event time; the within-adopter change across columns 1–2, not the gap with column 3, is the comparison of interest. "Number of households" reports the count of distinct households in the estimation sample for each group. The household count is the number of distinct households in that estimation sample. *Source:* Authors' calculations using the NielsenIQ Consumer Panel.

B The Estimation Sample

B.1 Adoption Cohorts and the Comparison Group

Table A8 reports how many adopters in each of the four half-yearly cohorts identify each event-time coefficient. Later coefficients rest on fewer cohorts, down to the 2024H1 cohort alone three half-years after adoption.

Table A8: Adoption Cohorts, Event Time, and What Identifies Each Coefficient

Adoption cohort (households)	$e = -5$	$e = -4$	$e = -3$	$e = -2$	$e = -1$	$e = 0$	$e = 1$	$e = 2$	$e = 3$
2024H1 (359)	—	—	—	**352**	355†	**350**	**348**	**315**	**315**
2024H2 (642)	—	—	**512**	**523**	632†	**635**	**565**	**565**	—
2025H1 (564)	—	**479**	**478**	**549**	549†	**559**	**558**	—	—
2025H2 (563)	**477**	**479**	**537**	**542**	556†	**563**	—	—	—
Never-treated (2,128)	11,285 household–half-year observations, none dated relative to adoption								
Adopters identifying	477	958	1,527	1,966	—	2,107	1,471	880	315

Notes: The table shows the cohort composition of the matched sample. Cells report household–half-year observations; e is the half-year relative to adoption. **Bold** cells receive a treatment dummy in the Sun and Abraham (2021) specification and therefore identify the event-time coefficients; † marks the omitted reference half-year ($e = -1$). Cells in neither category enter the comparison pool. Every adopting cohort has at least one pre-adoption half-year inside the window, so each receives treatment dummies and none is pooled into the comparison group; the 2,128 matched never-treated households are the only comparison units. The final row reports how many adopting households identify each event-time coefficient: the months 12–18 coefficient ($e = 2$) rests on the 2024H1 and 2024H2 cohorts (880 households), which is why the later-period estimates are imprecise. *Source:* Authors' calculations using the NielsenIQ Consumer Panel.

C Robustness of the Baseline Estimates

C.1 Alternative Staggered Difference-in-Differences Estimators

We re-estimate the calorie event study with five staggered difference-in-differences estimators (Figure A1). Panels (a) through (d) use the baseline estimation sample, the matched panel for 2023H1–2025H2 described in Appendix B.1. Panel (e) uses its balanced subpanel of households observed in all six half-years, because the efficient estimator requires a balanced panel. The five estimators differ in how they handle treatment effect heterogeneity across adoption cohorts and in the weights assigned to each cohort-time cell.

Panel (a): Sun and Abraham (2021). Our preferred specification. It uses an interaction-weighted regression that avoids the negative-weighting bias of standard TWFE under heterogeneous treatment effects. Each cohort's treatment effect is estimated separately and then aggregated using cohort-size weights.

Panel (b): Two-Way Fixed Effects (TWFE). The standard difference-in-differences regression with household and time fixed effects. Under staggered adoption with heterogeneous treatment effects, the TWFE estimator can assign negative weights to some cohort-period cells, producing biased estimates. We show it for comparison only.

Panel (c): Callaway and Sant'Anna (2021). A doubly robust estimator that computes cohort-specific average treatment effects $\text{ATT}(g,t)$ using never-treated households as the control group. It then aggregates across cohorts and time periods. It is robust to heterogeneous treatment effects by construction.

Panel (d): Borusyak, Jaravel, and Spiess (2024). An imputation-based estimator that fits a two-way fixed-effects model on pre-treatment observations to impute counterfactual outcomes for treated units in post-treatment periods.

Panel (e): Roth and Sant'Anna (2023). The minimum-variance linear estimator under staggered rollout designs. It was originally designed for settings with randomized treatment timing, and we apply it here to observational data using Neyman-robust standard errors ($\hat{\sigma}_{\text{Neyman}}$). However, the efficient estimator assumes treatment timing is as good as randomly assigned. That assumption fails in our observational setting, where GLP-1 adoption reflects underlying health conditions and physician guidance. The variance-optimal weights are designed for random timing, so under observational staggered adoption they are not guaranteed to produce the most informative estimates.

Its pre-adoption coefficients are flat. Its adoption-half-year decline is smaller, and it finds no decline afterward. Because panel (e) differs from the others in both its weights and its sample, we rely on

panels (a), (c), and (d) for inference.

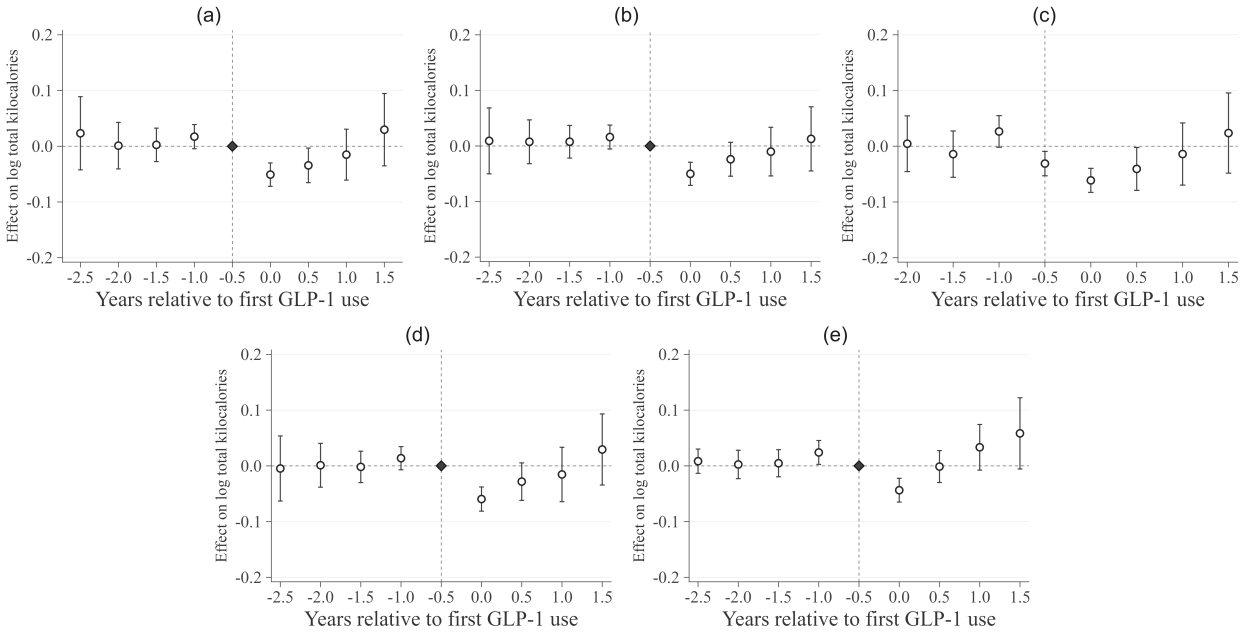


Figure A1: Event-study estimates of GLP-1 adoption on total calories purchased, across estimators

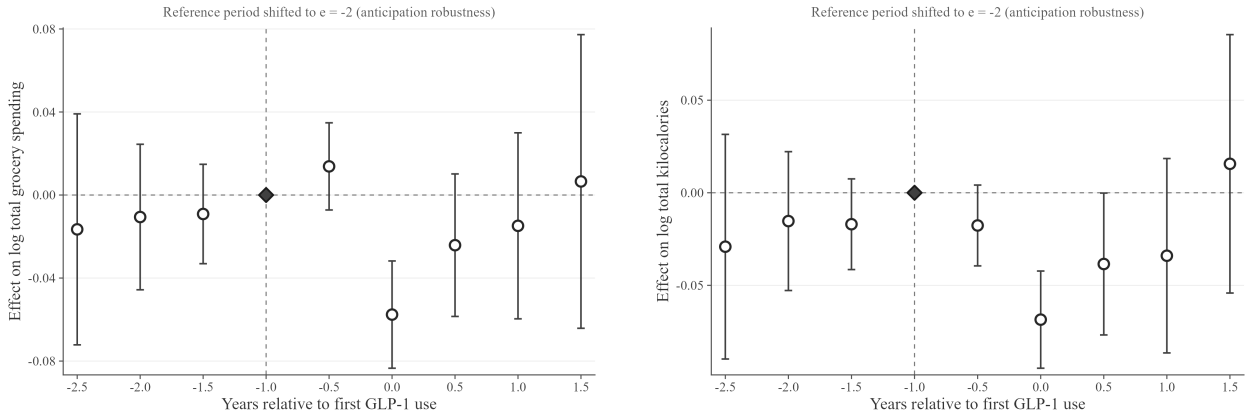
Notes: Each panel plots event-study coefficients (point estimates with 95 percent confidence intervals) from a different staggered difference-in-differences estimator. Panel (a) uses the Sun and Abraham (2021) interaction-weighted estimator; panel (b) uses the standard two-way fixed-effects (TWFE) regression; panel (c) uses the Callaway and Sant'Anna (2021) doubly robust estimator; panel (d) uses the Borusyak, Jaravel, and Spiess (2024) imputation estimator (BJS); panel (e) uses the Roth and Sant'Anna (2023) efficient estimator. The outcome is $\log(1 + y)$, where y is total calories purchased. The reference period is the half-year immediately before first GLP-1 use (year -0.5 on the horizontal axis, marked by the vertical dashed line). Panels (a)–(d) use the baseline estimation sample, the matched panel for 2023H1–2025H2 (Appendix B.1), which comprises 4,256 households (23,078 household–half-year observations); panel (e) uses its balanced subpanel of households observed in all six half-years, as the efficient estimator requires. In panel (c), the Callaway–Sant'Anna estimator compares each pre-adoption half-year with the one before it, so it reports a coefficient at year -0.5 rather than normalizing it to zero, and it cannot estimate year -2.5, whose preceding half-year falls before the sample window. Estimates at years 0 and 0.5 are broadly consistent across panels (a)–(d). Panel (e) shows a smaller adoption-half-year decline and no decline afterward. *Source:* Authors' calculations using the NielsenIQ Consumer Panel.

C.2 Anticipation

GLP-1 adoption is preceded by a physician visit and a filled prescription, so a household may begin changing what it buys before the half-year in which we first record use. Anticipation of this kind contaminates the normalized reference period. If spending has already fallen by the time we treat it as the baseline, every event-time coefficient is shifted by that amount. The estimated effect of adoption then understates the true one. The baseline specification cannot detect this, because the half-year immediately before adoption is the omitted category.

Figure A2 re-estimates the baseline specification for total grocery spending (panel a) and calories purchased (panel b) with the reference period moved back to year −1, which makes the coefficient at year −0.5 estimable. For spending, that coefficient is 1.4% and is not distinguishable from zero ($p = 0.20$). A joint Wald test does not reject that the estimated leads are zero ($p = 0.69$). The point estimate is positive, while anticipatory dieting would push spending down before adoption. The same holds for calories purchased (panel b). The coefficient at year −0.5 is −1.8% ($p = 0.11$), and the joint test does not reject that the leads are zero ($p = 0.28$).

The test disciplines anticipation at the half-year frequency and no finer. Consider a household that cuts its purchases within the same half-year in which it first reports GLP-1 use. In these data it is indistinguishable from one that cuts them because of the drug. We therefore read the estimates throughout as the effect of the adoption episode.



(a) Total grocery spending

(b) Total calories purchased

Figure A2: Anticipation robustness: shifting the reference period to year −1

Notes: Each panel replicates the baseline Sun and Abraham (2021) specification but shifts the reference period from year −0.5 to year −1, which allows the coefficient immediately preceding adoption (year −0.5) to be estimated. The outcome is $\log(1 + y)$, where y is total grocery spending (panel a) or total calories purchased (panel b). In neither panel is the year −0.5 coefficient distinguishable from zero, and the joint Wald test does not reject that all pre-treatment leads are zero (spending $p = 0.69$; calories $p = 0.28$). The estimation sample is the matched panel of 4,256 households (23,078 household–half-year observations). *Source:* Authors' calculations using the NielsenIQ Consumer Panel.

C.3 Sensitivity to Violations of Parallel Trends

Flat pre-adoption coefficients are evidence against differential trends, not proof of their absence. The leads are estimated with error, and a violation small enough to pass the pre-period test can still be large enough to account for a post-adoption estimate. Rambachan and Roth (2023) replace the test with a bound. Their relative-magnitudes restriction allows the post-adoption violation of

parallel trends to be as large as $\bar{M}$ times the largest violation observed between consecutive pre-adoption periods. It then traces out the confidence set that remains valid under that restriction as $\bar{M}$ grows.

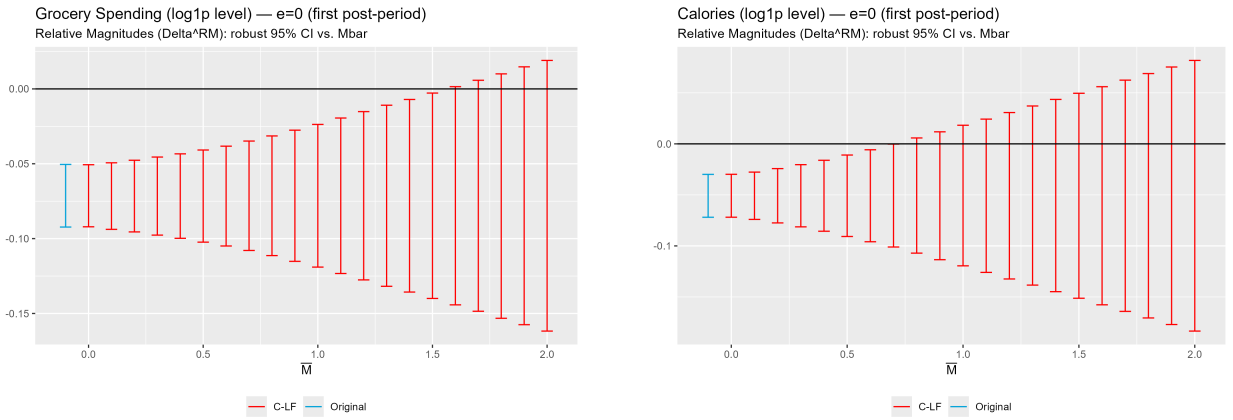
The breakdown value is the $\bar{M}$ at which the set first includes zero. It answers the question a reader wants answered: how much worse would the post-adoption world have to be than the pre-adoption one we can see? Rambachan and Roth recommend no particular $\bar{M}$, and none is standard across applications. The one benchmark that needs no further assumption is $\bar{M} = 1$, which holds the post-adoption violation to no more than the largest violation visible before adoption.

We run the exercise for the adoption half-year, the period on which we rest the causal reading (Figure A3). The spending decline clears that benchmark, surviving to $\bar{M} = 1.6$. A confounder timed to adoption would have to break parallel trends by more than the worst pre-adoption departure in order to explain it away. The calorie decline falls short, breaking down at $\bar{M} = 0.7$, so a post-adoption violation smaller than the worst pre-adoption one is enough. Calories are the more fragile of the two results, and we treat them accordingly.

The pooled post-adoption averages are weaker still. Averaging the adoption half-year together with the later ones, whose coefficients are estimated on a shrinking base, dilutes the effect without shrinking the pre-adoption violations it is measured against. The pooled spending effect breaks down at $\bar{M} = 0.6$. This is one reason we report coefficients period by period rather than leaning on the pooled effect.

C.4 Clustering

Matching pairs each adopting household with one never-adopting household, which raises the question of whether inference should be clustered on the matched pair rather than on the household. It makes almost no difference here. Re-clustering the baseline average effect at the pair level moves the standard error on total spending from 0.0122 to 0.0121. The standard error on calories is unchanged at four decimals, 0.0124 at the household level against 0.0124 at the pair level. Point estimates are identical by construction, since the level of clustering enters only the variance, and no significance conclusion changes.



(a) Total grocery spending

(b) Total calories purchased

Figure A3: Honest difference-in-differences sensitivity of the adoption-half-year effects

Notes: Each panel plots the Rambachan and Roth (2023) relative-magnitudes robust 95 percent confidence sets for the adoption-half-year effect (year 0) as a function of $\bar{M}$, the maximum allowed ratio of post-adoption to pre-adoption violations of parallel trends ($\bar{M} = 0$ reproduces the original confidence interval). The underlying event-study coefficients and their joint covariance are from the baseline Sun and Abraham (2021) specification with household and calendar half-year fixed effects and household-clustered standard errors. For spending (panel a), the robust confidence set excludes zero up to a breakdown value of $\bar{M} = 1.6$; for calories (panel b), the breakdown value is $\bar{M} = 0.7$. The estimation sample is the matched panel of 4,256 households (23,078 household–half-year observations). *Source:* Authors' calculations using the NielsenIQ Consumer Panel.

D Are the Food-Group Effects Equal?

Figure 3 reports the treatment effect separately for each of the eleven USDA ERS food groups, and the discussion in the text reads a ranking off that grid. The durable cut runs deepest for meat and beans. A grid of separately estimated coefficients cannot, by itself, support that reading. Each group is estimated on its own regression, so the estimates carry no covariance with one another. Nothing in Figure 3 tests whether the groups differ at all rather than scattering around a common effect.

Table A9 supplies the missing test. We stack the eleven food-group panels and estimate a single fully interacted Sun and Abraham (2021) specification (food group × adoption cohort × relative period). It carries household-by-food-group and half-year-by-food-group fixed effects and standard errors clustered at the household level.

Because the interaction is saturated, the stacked coefficients reproduce the group-by-group estimates plotted in Figure 3 to machine precision. The point estimates in the figure and in the test are the same numbers. What the stacked design adds is the covariance between them, which is exactly what a cross-group comparison requires and what separate regressions cannot provide. We then test the null that the effect is identical in all eleven groups, period by period and across the three post-adoption periods jointly.

The same covariance supports the Romano–Wolf correction reported in the main text. The usual implementation bootstraps the null distribution (Clarke et al. 2020). Here the stacked specification already delivers the joint covariance $\hat{\Sigma}$ of the aggregated coefficients, so we draw from it directly. We simulate $t^* \sim N(0, R)$, where R is the correlation matrix implied by $\hat{\Sigma}$, with 20,000 draws. The stepdown then runs within a single outcome, a family of 33 coefficients.

Equality is marginally rejected for grocery spending in the adoption half-year ($p = 0.06$) and rejected at the 1 percent level jointly across the three post-adoption periods ($p = 0.002$). For calories the rejection comes later. Equality is not rejected in the adoption half-year ($p = 0.29$) but is rejected two half-years out ($p = 0.001$). The joint test across the three periods also rejects ($p = 0.03$). The food-group heterogeneity discussed in the text therefore has statistical standing for both spending and calories.

Table A9: Are the Food-Group Effects Equal? Joint Tests across the Eleven ERS Groups

	F	**Restrictions**	p**-value**
Panel A. Grocery spending			
Adoption half-year	1.79	10	0.057*
+1 half-year	2.23	10	0.014**
+2 half-years	2.64	10	0.003***
All three periods jointly	1.93	30	0.002***
Panel B. Calories purchased			
Adoption half-year	1.20	10	0.286
+1 half-year	2.01	10	0.028**
+2 half-years	2.92	10	0.001***
All three periods jointly	1.54	30	0.031**

Notes: Each row tests the null that the treatment effect is the same in all eleven USDA ERS food groups, at the indicated post-adoption period or across all three periods jointly (30 restrictions). The estimates come from stacking the eleven food-group panels and estimating a fully interacted Sun and Abraham (2021) specification (food group × cohort × relative period), with household-by-food-group and half-year-by-food-group fixed effects and standard errors clustered at the household level. The stacked coefficients reproduce the group-by-group estimates plotted in Figure 3 to machine precision; what the stacked design adds is the covariance between them, which is what a cross-group comparison requires and what separate regressions cannot provide. F statistics are reported with $G - 1$ denominator degrees of freedom, where G is the number of households. *, **, and *** denote rejection of equality at the 10%, 5%, and 1% levels. *Source:* Authors' calculations using the NielsenIQ Consumer Panel.

E Nutrient Effects by Food Group

Table A10: Effect of GLP-1 Adoption on Total Nutrient Purchases

	ATT	Std. error
Calories (kcal)	-0.034***	(0.012)
Total Fat (g)	-0.057***	(0.013)
Saturated Fat (g)	-0.042***	(0.013)
Total Carbohydrate (g)	-0.012	(0.014)
Sugar (g)	0.008	(0.016)
Added Sugar (g)	0.016	(0.019)
Dietary Fiber (g)	-0.053***	(0.014)
Protein (g)	-0.046***	(0.012)
Sodium (g)	-0.040***	(0.013)
Observations	23,078	
Households	4,256	

Notes: Each row reports the average treatment effect on the treated (ATT) from the Sun and Abraham (2021) staggered difference-in-differences estimator. The outcome is $\log(1 + y)$, where y is the amount purchased of the nutrient named in the row, measured in the unit given beside its name and aggregated across all grocery purchases in each half-year. The sample consists of matched GLP-1 adopter and never-adopter households observed from 2023H1 through 2025H2, with household and calendar half-year fixed effects. Standard errors (in parentheses) are clustered at the household level. *Observations* is the number of household–half-year observations; *Households* is the number of distinct households (= number of clusters). Significance: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Figure A4 decomposes protein and sugar by food group.[3] The largest adoption-half-year protein declines are in table fats and oils (−12.2%), meat and beans (−9.7%), and gravies and condiments (−9.5%), joined by salty snacks, desserts and sweets, grains, and vegetables. Each falls significantly in the adoption half-year.

Sugar behaves differently: its basket-level change is indistinguishable from zero (Table A10). All eleven groups carry negative adoption-half-year point estimates, and table fats and oils (−14.1%), meat and beans (−10.0%), salty snacks, and grains are individually significant at the 5 percent level.

The later periods sort the groups into distinct patterns. Meat and beans is the one group where both cuts persist and deepen, reaching −12.6% for protein and −18.5% for sugar two half-years out (both significant). The other adoption-half-year declines do not persist: salty snacks fades toward zero, while grains and sugar in table fats and oils turn positive, though not significantly. Working against the average, the discretionary and sweet groups drift upward. Desserts and sweets carry positive point estimates two half-years out for both protein (+5.2%) and sugar, though neither is

[3] As in the main text, this decomposition is exploratory. The p-values reported below are unadjusted for multiple testing, and we read the patterns as descriptive rather than confirmatory.

significant. Beverages also carry positive later-period point estimates. The rebound mirrors the discretionary-spending pattern in Figure 3 and is consistent with substitution back toward palatable foods as the initial appetite effect fades.

Across all four nutrients the reductions that persist are in meat and beans, the staple that also carries the most durable spending response, while discretionary groups recover (Figure A5). Sodium falls broadly and cleanly in the adoption half-year, across salty snacks (−7.2%), meat and beans (−8.1%), grains (−6.1%, all three $p < 0.01$), vegetables, and gravies and condiments. Almost no group moves against the trend. Fiber also falls broadly in the adoption half-year, across salty snacks (−8.5%), meat and beans (−9.6%), desserts and sweets, gravies and condiments, grains, and vegetables. Only beverages carry a positive point estimate (+4.8%), and it is not significant.

The fiber and sodium dynamics again single out meat and beans, where both are deepest two half-years after adoption (−9.1% for sodium and −18.0% for fiber). Grains lose significance after the adoption half-year, and salty snacks one half-year later.

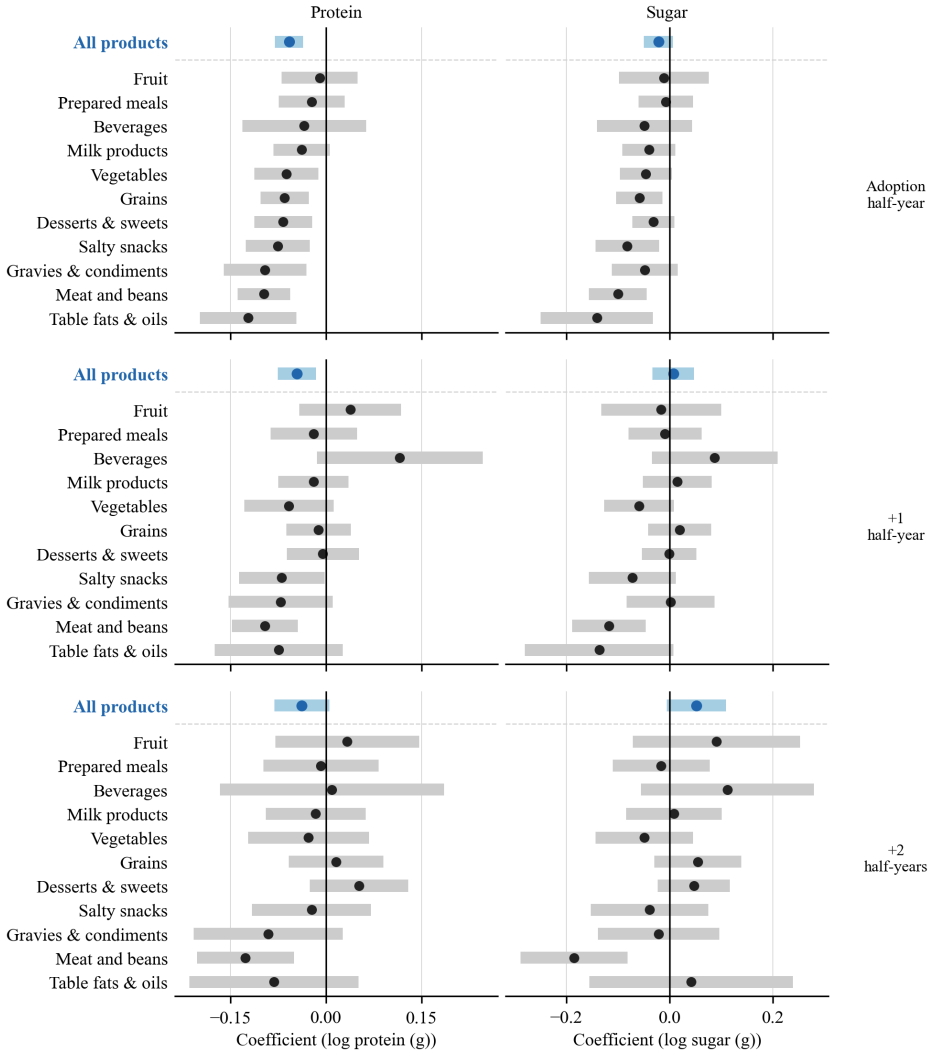


Figure A4: Effect of GLP-1 adoption on protein and sugar purchased by food group, by post-adoption period

Notes: Each point shows the Sun and Abraham (2021) staggered difference-in-differences coefficient for log(1 + grams) of protein (left column) and sugar (right column) purchased within each food group, in the adoption half-year and the two half-years that follow (rows). The eleven food groups follow the USDA Economic Research Service (ERS) food-group classification (USDA-ERS 2016) and are ordered by the adoption-half-year protein coefficient (ascending). The highlighted "All products" row at the top of each panel reports the corresponding whole-basket nutrient coefficient (Table A10) for the same outcome and period, as a benchmark. The sample consists of GLP-1 adopter and never-adopter households observed from 2023H1 through 2025H2, with household and calendar half-year fixed effects; each food-group panel is filled to the complete grid of 4,256 households, with half-years in which a household buys nothing in a group entered as zero, so every per-group regression uses the same sample. Horizontal bars denote 95 percent confidence intervals. Standard errors are clustered at the household level. *Source:* Authors' calculations using the NielsenIQ Consumer Panel.

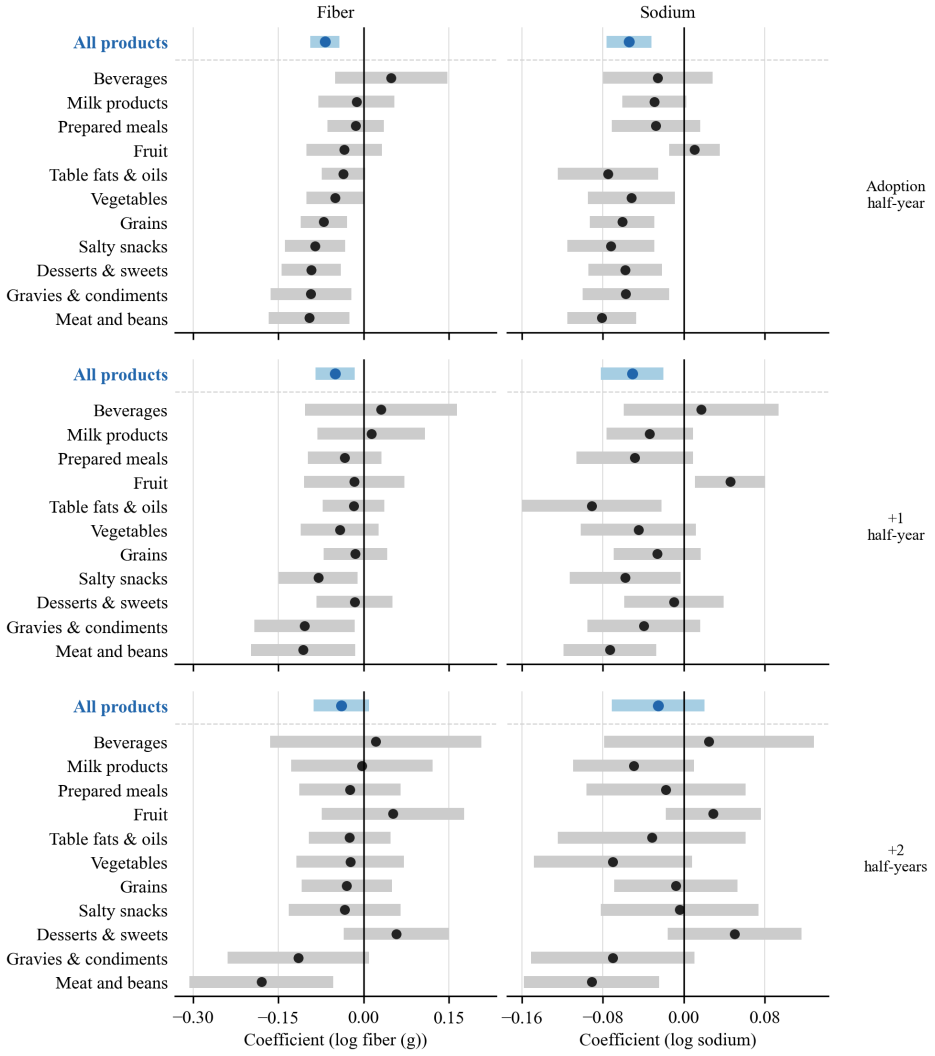


Figure A5: Effect of GLP-1 adoption on dietary fiber and sodium purchased by food group, by post-adoption period

Notes: Each point shows the Sun and Abraham (2021) staggered difference-in-differences coefficient for log(1 + grams) of dietary fiber (left column) and sodium (right column) purchased within each food group, in the adoption half-year and the two half-years that follow (rows). Sodium is measured in grams, not milligrams. The eleven food groups follow the USDA Economic Research Service (ERS) food-group classification (USDA-ERS 2016) and are ordered by the adoption-half-year fiber coefficient (ascending). The highlighted "All products" row at the top of each panel reports the corresponding whole-basket nutrient coefficient (Table A10) for the same outcome and period, as a benchmark. The sample consists of GLP-1 adopter and never-adopter households observed from 2023H1 through 2025H2, with household and calendar half-year fixed effects; each food-group panel is filled to the complete grid of 4,256 households, with half-years in which a household buys nothing in a group entered as zero, so every per-group regression uses the same sample. Horizontal bars denote 95 percent confidence intervals. Standard errors are clustered at the household level. *Source:* Authors' calculations using the NielsenIQ Consumer Panel.